

Objective Sublevels and Central-Path Hessian Conditioning

Abstract

We characterize the equality-reduced primal barrier Hessian by the geometry of objective sublevels in a fixed Euclidean metric. For a compact convex feasible set, a nonconstant linear objective, and any finite-parameter self-concordant barrier, the unit Dikin ellipsoid at a central point of primal gap g is comparable with the difference body $L(g) - L(g)$. This yields a Loewner comparison between different barriers at equal gap and the matching law $\kappa(g) = \Theta((\text{diam } L(g)/g)^2)$. Explicit constants extend the results to points with Newton decrement below one and to uniformly bounded families of barrier parameters. Directional widths determine every ordered spectral scale. We derive conditioning regimes, a uniform near-degeneracy plateau, and four exact leading eigenvalue constants for the reduced log-Hessian of a classical 3×3 semidefinite construction. A paired section shows that singularity degree does not determine the conditioning exponent. For linear programs, every fixed barrier has at most two spectral scales; its weak eigenspace approaches the optimal-face tangent at rate $O(g)$, improved to $O(g^2)$ for the logarithmic barrier. A globally certified example attains the general rate and separates small discarded residual from accurate Newton direction recovery. We state the resulting clustered-CG consequence and the additional assumptions needed for quantum solver interpretations. Reproducible computations include exact rational input certificates and explicit precision screens.

Keywords: self-concordant barriers; central path; Hessian conditioning; objective sublevels; semidefinite programming; spectral clustering.

1 Introduction

The Newton equations of an interior-point method can become increasingly ill-conditioned as the requested objective accuracy improves. This behavior is often described through the particular logarithmic barrier or through a chosen primal–dual linear system. A more basic question is which part of the behavior is already forced by the feasible set and the objective. If a different self-concordant barrier follows a different central path, can it improve the accuracy exponent of the primal barrier Hessian? Which geometric features determine the individual eigenvalue scales? These questions matter both when choosing a barrier and when interpreting the condition-number dependence of a linear-system solver.

We answer them for the equality-reduced primal Hessian in a fixed external metric. Let P be a compact convex feasible set with nonempty relative interior and let a linear objective be nonconstant on its affine hull. For primal objective gap g , let $L(g)$ be the feasible sublevel set and $D(g) = \text{diam } L(g)$. If $\hat{H}_F(g)$ denotes the unscaled barrier Hessian at a central point of gap g , our main conclusion is

$$\kappa(\hat{H}_F(g)) = \Theta\left(\frac{D(g)^2}{g^2}\right) \quad (g \downarrow 0) \quad (1)$$

for every fixed finite-parameter self-concordant barrier F . Constants may depend on the fixed instance, metric, and barrier parameter, but not on g . Thus changing the barrier can change constants and the location of the center, while the accuracy exponent is determined by objective

geometry. The statement also has a uniform version for bounded barrier parameters and centrality residuals bounded uniformly below one. Compact objective sublevels suffice for a localized version on an unbounded feasible set.

The underlying comparison is stronger than (1). The symmetric difference body $K_g = L(g) - L(g)$ contains the unit tangent Dikin ellipsoid E_F at an approximately central point of observed gap g , and is contained in $2CE_F$, where

$$C = \frac{\nu + 2\sqrt{\nu} + \rho}{1 - \rho}, \quad 0 \leq \rho < 1.$$

Here ν is the barrier parameter and ρ is the Newton decrement of $F + c^\top x/\mu$, measured using the unscaled Hessian. Consequently, Hessians of different barriers at the same primal gap are comparable in Loewner order. This controls all quadratic forms and ordered eigenvalues, even when the two centers differ. A homothetic interior ball gives the objective-scale thickness of K_g ; its diameter then supplies the matching condition-number law.

1.1 Relationship to prior work

The geometry of self-concordant barriers is classical (Nesterov and Nemirovskii, 1994; Renegar, 2001; Nesterov, 2018). Dikin containment, semiboundedness, and asymmetric containment are the ingredients used here; we give the short proofs needed for our constants. In particular, Nesterov and Nemirovskii (1994, Proposition 2.3.2(iii), equation (2.3.9)) compares the local Hessian norm with the symmetric chord radius at a fixed interior point. It therefore already compares different barriers' Hessians at that *same point*. Our equal-gap result compares Hessians at generally different centers, through one objective difference body. The connection between objective-level geometry and conditioning also has substantial precedent. Freund (2003) studies primal–dual level-set geometry. More directly, Xiong and Freund (2024, Section 5.1, Fact 5.2 and Remark 5.1) use central-path ellipsoids to relate the diameter and conic radius of primal–dual sublevels to Hessian eigenvalues. They use these geometric relations to analyze rescaling for PDHG. Their work is a direct antecedent to our geometric method. We instead compare the tangent primal Hessian with a center-independent difference body at a prescribed primal gap, and obtain the matching law (1) and comparison between barriers.

Classical matrix-conditioning analyses are equally relevant. Renegar (1996) relates the conditioning of barrier linear equations to problem conditioning and conjugate-gradient solution. Peña (2001, 2002) develops a primal–dual condition perspective and implicit-barrier methods. In particular, Peña (2002, Proposition 3.2) parameterizes the central path by primal objective gap, and his Section 3.3 bounds the condition number of a Schur matrix $A\nabla^2 F(x)^{-1}A^\top$ by an inverse-square gap rate under its stated assumptions. Accordingly, neither objective-gap parameterization nor a generic inverse-square upper bound is a novelty claim here. The operator, the matching dependence on $D(g)$, and the comparison between barriers must be specified to identify the additional result.

Spectral splitting of logarithmic-barrier Hessians and its numerical consequences were studied by Wright (1994, 1998); Forsgren et al. (2002) provides a broader account of interior methods, and Wright and Orban (2002) treats degenerate nonlinear programs. For linear programming, centered endpoint limits and their analytic properties are classical (Adler and Monteiro, 1991; Güler, 1994; Halická, 1999). Our scaled-Hessian limit at a unique LP optimum, including a degenerate vertex, is explicitly presented as a corollary of this theory. Duistermaat (2001) analyzes boundary asymptotics of the Hessian Riemannian structure under additional smoothness assumptions; his Remark 2.3 gives oscillatory self-concordant perturbations. Thus oscillation of barrier derivatives is also an established phenomenon. The additional LP results transfer every ordered spectral scale to arbitrary

finite-parameter barriers and control the weak eigenspace relative to the fixed optimal-face tangent. This gives an $O(g)$ weak component of the exact Newton forcing, improved to $O(g^2)$ for the canonical logarithmic barrier. An explicitly certified barrier shows that the general rate is sharp.

For semidefinite programming, error bounds and facial reduction connect singularity degree with fractional powers (Sturm, 2000; Drusvyatskiy and Wolkowicz, 2017; Lourenço, 2021). We use a trace-normalized version of Sturm’s classical 3×3 construction, rather than claim a new source of quarter-power error geometry. SDP spectral analyses also predate this work: Alizadeh et al. (1998) studies a primal–dual Schur matrix, Wei and Wolkowicz (2010) investigates instances with failure of strict complementarity, and Sremac et al. (2021) relates singularity degree to primal and dual spectral rates on an external regularization path. Our operator is the Hessian compressed to the fixed equality tangent, and our affine constraints do not vary with the path parameter. We compute its four leading eigenvalue constants and exhibit a section with the same optimality-system singularity degree but a different conditioning exponent. The detailed comparison in section 7 distinguishes the relevant paths and matrices.

Finally, spectral condition numbers alone do not determine solver cost. Cluster-sensitive conjugate-gradient analysis is classical (Axelsson, 1994); our explicit polynomial bound states the precise exact-arithmetic energy-error consequence needed here. Quantum linear-system results additionally depend on access, normalization, right-hand-side preparation, and output requirements (Gilyén et al., 2019; Orsucci and Dunjko, 2021). These distinctions remain necessary in quantum interior-point methods, including the positive LP results of Apers and Gribling (2026). We give no end-to-end quantum lower bound from the geometric conditioning law.

1.2 Contributions and organization

The principal developments are the following.

1. *Comparison at equal objective accuracy.* Theorems 3.2, 3.3 and 4.1 prove the explicit difference-body containment, equal-gap Loewner comparison, matching diameter law, and full-spectrum width bounds. The estimates hold at an iterate’s own gap for every residual below one, and are uniform over families with bounded barrier parameters and centrality residuals bounded uniformly below one. To the best of our knowledge, this equal-gap comparison between arbitrary finite-parameter barriers, with these residual-dependent bounds and the resulting matching diameter characterization, has not previously been stated in this form.
2. *Sharp geometric and spectral examples.* The law gives bounded conditioning at every unique polytope optimum, including degenerate vertices, and inverse-square conditioning at a positive-dimensional optimal face. A near-degenerate simplex has a uniform transition between a large plateau and an inverse-square regime. For the trace-normalized classical SDP example, theorem 7.2 derives all four reduced-Hessian scales and constants. Its paired section proves that singularity degree alone does not determine the exact conditioning exponent. The classical endpoint and error-bound ingredients are identified separately from these explicit calculations and their all-barrier consequences.
3. *Newton forcing and the limits of a condition-number analysis.* Theorem 8.1 and proposition 8.2 establish the general $O(g)$ and canonical $O(g^2)$ weak-projector rates and the corresponding exact-forcing bounds. The sharp example has a globally verified finite barrier parameter. It also shows that discarding a vanishing relative right-hand-side component can lose asymptotically all Euclidean solution mass. The specific rate and solution-error calculation are the contribution; qualitative nonconvergence of central paths has prior examples (Bolte and Pauwels, 2022).

Sections 2 to 4 develop the general theory; sections 5 to 7 give classifications and exact examples. Sections 8 and 9 connect spectra to Newton forcing and linear solves, and section 10 identifies the metric and barrier-parameter boundaries. Section 11 supplies reproducible numerical illustrations with exact input certificates and explicit precision screens. All asymptotic claims used in interpreting those computations are proved in the preceding sections.

2 The objective gap and the reduced Hessian

Throughout the main geometric results, P is a compact convex subset of a finite-dimensional Euclidean space, with affine hull $x^0 + \mathcal{V}$ and relative interior $P^\circ = \text{ri } P$. The tangent space $\mathcal{V}$ has dimension $d \geq 1$. We minimize a linear functional $\langle c, x \rangle$ that is nonconstant on P ; equivalently, its orthogonal projection $c_{\mathcal{V}}$ onto $\mathcal{V}$ is nonzero. Set

$$\text{OPT} = \min_{x \in P} \langle c, x \rangle, \quad \Phi = \{x \in P : \langle c, x \rangle = \text{OPT}\}, \quad (2)$$

$$g(x) = \langle c, x \rangle - \text{OPT}, \quad \Delta = \max_{x \in P} g(x) > 0. \quad (3)$$

For $0 \leq g \leq \Delta$, define

$$L(g) = \{y \in P : g(y) \leq g\}, \quad D(g) = \text{diam } L(g), \quad K_g = L(g) - L(g) \subset \mathcal{V}. \quad (4)$$

The difference body K_g consists of all chords of the sublevel set, translated to the origin. It retains directional information that the diameter alone discards. For an interior point x , also put

$$\ell(x) = \max_{y \in L(g(x))} \|y - x\|, \quad \frac{D(g(x))}{2} \leq \ell(x) \leq D(g(x)). \quad (5)$$

The lower inequality follows by applying the triangle inequality to a pair of points attaining the diameter.

Definition 2.1 (Barrier and metric). A ν -self-concordant barrier for P , with $\nu > 0$, is a convex C^3 function $F : P^\circ \rightarrow \mathbb{R}$ that tends to $+\infty$ at every relative boundary point, has positive-definite tangent Hessian, and satisfies

$$|D^3F(x)[h, h, h]| \leq 2(D^2F(x)[h, h])^{3/2}, \quad (6)$$

$$|DF(x)[h]| \leq \sqrt{\nu}(D^2F(x)[h, h])^{1/2} \quad (h \in \mathcal{V}). \quad (7)$$

All gradients and Hessians below are relative to $x^0 + \mathcal{V}$. Write $H_F(x)$ for the self-adjoint operator on $\mathcal{V}$ representing $D^2F(x)$, and define

$$\|h\|_x = \sqrt{\langle H_F(x)h, h \rangle}, \quad \|v\|_x^* = \sqrt{\langle H_F(x)^{-1}v, v \rangle}, \quad E_F(x) = \{h \in \mathcal{V} : \|h\|_x \leq 1\}. \quad (8)$$

The condition number is $\kappa_F(x) = \lambda_{\max}(H_F(x))/\lambda_{\min}(H_F(x))$.

If W is an orthonormal basis matrix for $\mathcal{V}$, the matrix of this operator is $W^\top \nabla^2 F(x) W$. Consequently its eigenvalues are independent of the chosen orthonormal basis. The external metric is fixed throughout; for symmetric matrix variables it is the Frobenius metric, and for several blocks it is the product Frobenius metric. We use the Hessian of F , rather than μF . Multiplication by μ leaves its condition number unchanged but changes every eigenvalue scale. A change of feasible representation or a nonorthogonal change of coordinates requires its own metric analysis.

2.1 Two standard barrier facts, with proofs

The following facts are standard self-concordant-barrier geometry (Nesterov and Nemirovskii, 1994; Renegar, 2001). We give their short one-dimensional proofs to fix the normalization and the treatment of boundary points.

Lemma 2.2 (Dikin containment and semiboundedness). *For $x \in P^\circ$,*

$$x + \text{int}_{\mathcal{V}} E_F(x) \subset P^\circ, \quad x + E_F(x) \subset P, \quad (9)$$

and for every $y \in P$,

$$\langle \nabla F(x), y - x \rangle \leq \nu. \quad (10)$$

In particular, at every interior point, whether central or not,

$$\|c_{\mathcal{V}}\|_x^* \leq g(x). \quad (11)$$

Proof. Fix $h \in \mathcal{V}$ with $\|h\|_x = 1$, and write $\varphi(t) = F(x + th)$ for $0 \leq t < R$, where R is the first positive exit time from P° . Self-concordance gives

$$\left| \frac{d}{dt} \frac{1}{\sqrt{\varphi''(t)}} \right| \leq 1. \quad (12)$$

Thus $\varphi''(t) \leq (1 - t)^{-2}$ for $0 \leq t < \min\{R, 1\}$. If $R < 1$, integration twice bounds $\varphi(t)$ as $t \uparrow R$, contradicting boundary divergence. Hence $R \geq 1$. This proves the open containment, and closed containment follows by taking limits in the closed set P .

To prove (10), the case $y = x$ is immediate. Otherwise let $\psi(t) = F(x + t(y - x))$, $0 \leq t < 1$. These points are interior by convexity and the relative interior assumption on x . If $\psi'(0) \leq 0$, the claim is immediate. If $\psi'(0) > 0$, convexity and (7) give

$$\psi'(t) > 0, \quad \psi''(t) \geq \frac{\psi'(t)^2}{\nu}, \quad \frac{1}{\psi'(s)} \leq \frac{1}{\psi'(0)} - \frac{s}{\nu} \quad (0 \leq s < 1).$$

The left side is positive. Letting $s \uparrow 1$ proves $\psi'(0) \leq \nu$, including boundary choices of y . Finally, minimizing the linear objective over the closed Dikin ellipsoid gives $\langle c, x \rangle - \|c_{\mathcal{V}}\|_x^* \geq \text{OPT}$. $\square$

2.2 Existence and objective-gap parameterization

For $\mu > 0$, the central point is

$$x_F(\mu) = \arg \min_{x \in P^\circ} \left(F(x) + \frac{\langle c, x \rangle}{\mu} \right), \quad \nabla F(x_F(\mu)) = -\frac{c_{\mathcal{V}}}{\mu}. \quad (13)$$

The gap $g_F(\mu) = g(x_F(\mu))$ is a *primal objective gap*. Even for a canonical logarithmic barrier, it need not equal the primal–dual gap $\nu\mu$.

The use of objective value to parameterize central paths is classical. In particular, the bounds in the following proposition agree with Peña (2002, Proposition 3.2). The proof here uses only Dikin support and differentiation of centrality.

Proposition 2.3 (The attained gap interval). *The analytic center $x_F^{\text{ac}} = \arg \min F$ and all points $x_F(\mu)$ exist and are unique. Put $a_F = g(x_F^{\text{ac}}) \in (0, \Delta)$. Then g_F is continuously differentiable, strictly increasing, and maps $(0, \infty)$ bijectively onto $(0, a_F)$. Moreover,*

$$g'_F(\mu) = \frac{\|c_V\|_{x_F(\mu)}^{*2}}{\mu^2} > 0, \quad (14)$$

$$\frac{\mu a_F}{\mu + a_F} \leq g_F(\mu) \leq \nu \mu. \quad (15)$$

Thus $g_F(\mu) = \Theta(\mu)$ as $\mu \downarrow 0$ for every fixed barrier.

Proof. A supporting affine function of F bounds it below on compact P . Boundary divergence then makes its bounded sublevel sets compact subsets of P° . The same holds after adding any linear function. Existence follows, and the positive-definite Hessian gives strict convexity and uniqueness. The implicit function theorem applied to (13) gives $x'_F(\mu) = H_F(x_F(\mu))^{-1} c_V / \mu^2$, proving (14).

For any $x^* \in \Phi$, centrality and semiboundedness give

$$g_F(\mu) = \mu \langle \nabla F(x_F(\mu)), x^* - x_F(\mu) \rangle \leq \nu \mu.$$

Hence $g_F(\mu) \rightarrow 0$ as $\mu \downarrow 0$. For $\mu \rightarrow \infty$, optimality gives $F(x_F(\mu)) \leq F(x_F^{\text{ac}}) + \Delta/\mu$. Compactness of a fixed barrier sublevel and uniqueness of its minimizer imply $x_F(\mu) \rightarrow x_F^{\text{ac}}$. An interior point cannot minimize or maximize a nonconstant linear functional, so $0 < a_F < \Delta$. Strict monotonicity proves the claimed range.

By (11) and (14), $(1/g_F)' \geq -1/\mu^2$. Integration from μ to M and then $M \rightarrow \infty$ yield $1/g_F(\mu) \leq 1/a_F + 1/\mu$, which is the remaining inequality. $\square$

Let $\mu_F : (0, a_F) \rightarrow (0, \infty)$ be the inverse of g_F . We distinguish the gap-parameterized path and its Hessian and condition number by writing

$$\hat{x}_F(g) = x_F(\mu_F(g)), \quad \hat{H}_F(g) = H_F(\hat{x}_F(g)), \quad \hat{\kappa}_F(g) = \kappa_F(\hat{x}_F(g)). \quad (16)$$

Equal-gap comparisons of two fixed barriers are therefore meaningful on the common interval $(0, \min\{a_F, a_G\})$. Equal values of g generally correspond to different values of μ . This distinction is essential when the Hessians themselves, rather than only their condition numbers, are compared.

3 The difference body determines the Hessian

The geometric mechanism has two parts. The objective-decreasing half of a Dikin ellipsoid lies in the current sublevel set. Conversely, centrality places the entire sublevel set in a bounded dilation of that ellipsoid. Passing to differences removes the center and makes comparison between barriers possible. The same argument tolerates a nonzero centrality residual, measured at the point whose gap is used.

The containment ingredients are classical. The fixed-point symmetric-chord comparison of Nesterov and Nemirovskii (1994, Proposition 2.3.2(iii)) already compares Hessians of different barriers at one interior point. Here equal objective gap allows the compared points themselves to differ. Their use to relate sublevel geometry and Hessian eigenvalues is also established: for primal–dual conic sublevels, Xiong and Freund (2024, Fact 5.2 and Remark 5.1) bound diameters and conic radii using central-path ellipsoids, and use this geometry to analyze rescaling. Here we work with the tangent primal Hessian and a prescribed primal gap. The additional conclusions proved below are a difference-body comparison, a Loewner comparison between different barriers at equal gap, and a matching diameter law for every fixed barrier. The inverse-square upper edge and objective-gap parameterization are not claimed as new; see Peña (2002); Renegar (1996).

3.1 Containment with an explicit centrality residual

For $x \in P^\circ$ and $\mu > 0$, define the tangent Newton decrement

$$\rho_F(x; \mu) = \|\nabla F(x) + c_V/\mu\|_x^*. \quad (17)$$

This is the decrement of $F + \langle c, \cdot \rangle / \mu$, whose Hessian is $H_F(x)$. We call x approximately central when this residual is at most some $\rho < 1$. Exact centrality is the case $\rho = 0$. Define

$$C(\nu, \rho) = \frac{\nu + 2\sqrt{\nu} + \rho}{1 - \rho}, \quad C_\nu = C(\nu, 0) = \nu + 2\sqrt{\nu}. \quad (18)$$

Lemma 3.1 (One-sided containment). *Suppose $x \in P^\circ$, $y \in P$, and*

$$\langle \nabla F(x), y - x \rangle \geq -\rho \|y - x\|_x, \quad 0 \leq \rho < 1. \quad (19)$$

Then $\|y - x\|_x \leq C(\nu, \rho)$. Consequently, if $\rho_F(x; \mu) \leq \rho < 1$ for some $\mu > 0$ and $g = g(x)$, then

$$L(g) \subset x + C(\nu, \rho)E_F(x). \quad (20)$$

Proof. The assertion is immediate if $y = x$. Otherwise set $R = \|y - x\|_x$, $h = (y - x)/R$, and $\varphi(t) = F(x + th)$ for $0 \leq t < R$. The segment is interior up to its endpoint, even when y is on the boundary. Now $\varphi''(0) = 1$ and $\varphi'(0) \geq -\rho$. The other direction of (12) gives

$$\varphi''(t) \geq (1 + t)^{-2}, \quad \varphi'(t) \geq \frac{t}{1 + t} - \rho = \frac{(1 - \rho)t - \rho}{1 + t}. \quad (21)$$

Set $t_0 = (\sqrt{\nu} + \rho)/(1 - \rho)$. If $R \leq t_0$, the claimed bound follows because $t_0 < C(\nu, \rho)$. If $R > t_0$, semiboundedness at $x + t_0 h$, with endpoint y , gives

$$\varphi'(t_0)(R - t_0) \leq \nu, \quad \varphi'(t_0) \geq \frac{\sqrt{\nu}}{1 + t_0} > 0.$$

Therefore

$$R \leq t_0 + \sqrt{\nu}(1 + t_0) = \frac{\nu + 2\sqrt{\nu} + \rho}{1 - \rho}.$$

All estimates use derivatives only at interior points, so no derivative at y is required.

For the consequence, let $e = \nabla F(x) + c_V/\mu$. For $y \in L(g)$, $\langle c, y - x \rangle \leq 0$, and hence

$$\langle \nabla F(x), y - x \rangle = \langle e, y - x \rangle - \frac{\langle c, y - x \rangle}{\mu} \geq -\|e\|_x^* \|y - x\|_x.$$

Apply the first assertion. □

At $\rho = 0$ this is the usual asymmetric containment with constant $\nu + 2\sqrt{\nu}$; this constant is also used by [Xiong and Freund \(2024, Fact 5.2\)](#), citing [Nesterov \(2018, Theorem 5.3.8\)](#). The proof above supplies the stated extension for every residual below one. It does not require finding an exact center near x , and its sublevel is evaluated at the observed gap $g(x)$.

Theorem 3.2 (Difference-body and equal-gap comparisons). *Let $x \in P^\circ$ have gap g and satisfy $\rho_F(x; \mu) \leq \rho < 1$ for some $\mu > 0$. With $C = C(\nu_F, \rho)$ and $E = E_F(x)$,*

$$E \subset K_g \subset 2CE. \quad (22)$$

In particular, suppose barriers F, G have points x, y of the same gap g , with residual bounds $\rho_F, \rho_G < 1$, respectively. Write $H_F = H_F(x)$, $H_G = H_G(y)$, and $C_F = C(\nu_F, \rho_F)$, $C_G = C(\nu_G, \rho_G)$. Then, on the fixed tangent space,

$$\frac{1}{4C_F^2}H_F \preceq H_G \preceq 4C_G^2H_F. \quad (23)$$

The ordered eigenvalues obey the same inequalities. The condition numbers satisfy

$$\frac{\kappa_F(x)}{16C_F^2C_G^2} \leq \kappa_G(y) \leq 16C_F^2C_G^2\kappa_F(x). \quad (24)$$

Proof. For $h \in E$, choose $\sigma \in \{-1, 1\}$ so that $\langle c, \sigma h \rangle \leq 0$. Closed Dikin containment gives $x + \sigma h \in L(g)$, while $x \in L(g)$. Thus $\sigma h \in K_g$. The difference body is symmetric, so $h \in K_g$. This inclusion requires no centrality. The reverse inclusion follows from (20): the difference of two vectors of local norm at most C has local norm at most $2C$.

At equal gap, $E_F(x) \subset K_g \subset 2C_GE_G(y)$. Applying this to an arbitrary nonzero vector normalized in the H_F norm proves $H_G \preceq 4C_G^2H_F$. Interchanging F, G proves the other inequality. The eigenvalue inequalities follow from the variational principle, and their ratios give (24). $\square$

The center itself can move substantially when the barrier changes. Nevertheless, its Hessian ellipsoid has the same shape up to the displayed factors, because both ellipsoids approximate the same set of chords. This compares more than condition numbers: every quadratic form and every ordered eigenvalue are controlled. Exact centers are covered on their common attained gap interval from [proposition 2.3](#).

3.2 A matching conditioning law for every barrier

The difference-body comparison already characterizes conditioning by an aspect ratio without an additional instance constant. To express it using objective accuracy, fix a closed relative ball

$$z + r\mathbb{B}_{\mathcal{V}} \subset P, \quad r > 0, \quad (25)$$

where $\mathbb{B}_{\mathcal{V}}$ is the Euclidean unit ball in $\mathcal{V}$. Such a ball exists because P° is nonempty. Its radius and the objective range Δ measure the fixed instance, independently of the barrier.

Theorem 3.3 (Aspect ratio, spectral edges, and the diameter law). *Under the hypotheses of [theorem 3.2](#), let*

$$r_g = \max\{s \geq 0 : s\mathbb{B}_{\mathcal{V}} \subset K_g\}.$$

Then $r_g > 0$ and

$$\left(\frac{D(g)}{2Cr_g}\right)^2 \leq \kappa_F(x) \leq \left(\frac{2CD(g)}{r_g}\right)^2. \quad (26)$$

More explicitly,

$$\ell(x)^{-2} \leq \lambda_{\min}(H_F(x)) \leq C^2\ell(x)^{-2}, \quad (27)$$

$$\frac{\|c_{\mathcal{V}}\|^2}{g^2} \leq \lambda_{\max}(H_F(x)) \leq \left(\frac{C\Delta}{r_g}\right)^2, \quad (28)$$

$$\left(\frac{\ell(x)\|c_{\mathcal{V}}\|}{Cg}\right)^2 \leq \kappa_F(x) \leq \left(\frac{C\Delta\ell(x)}{r_g}\right)^2. \quad (29)$$

Consequently,

$$\frac{\|c_{\mathcal{V}}\|^2}{4C^2} \left(\frac{D(g)}{g} \right)^2 \leq \kappa_F(x) \leq \frac{C^2 \Delta^2}{r^2} \left(\frac{D(g)}{g} \right)^2. \quad (30)$$

The lower bounds for $\lambda_{\min}$ and $\lambda_{\max}$ hold at every interior point, without any centrality condition.

Proof. The Euclidean circumradius of K_g is $D(g)$. An ellipsoid with Hessian H has circumradius $\lambda_{\min}(H)^{-1/2}$ and inradius $\lambda_{\max}(H)^{-1/2}$. Apply both facts to (22) and divide the resulting inequalities to obtain (26).

For any Euclidean unit $h \in \mathcal{V}$, the vector $h/\sqrt{\langle H_F(x)h, h \rangle}$ is in $E_F(x)$. Choosing its objective-decreasing sign, as in the proof of theorem 3.2, shows that its Euclidean length is at most $\ell(x)$. Thus every unit-direction Rayleigh quotient is at least $\ell(x)^{-2}$. Conversely, choose $y \in L(g)$ with $\|y - x\| = \ell(x) > 0$. By (20), the Rayleigh quotient in direction $(y - x)/\ell(x)$ is at most $C^2/\ell(x)^2$. This proves (27).

The lower bound for $\lambda_{\max}$ follows from (11), since

$$g^2 \geq \langle H_F(x)^{-1} c_{\mathcal{V}}, c_{\mathcal{V}} \rangle \geq \frac{\|c_{\mathcal{V}}\|^2}{\lambda_{\max}(H_F(x))}.$$

For the upper bound choose $x^* \in \Phi$. Because $0 < g < \Delta$, convexity and linearity give the homothetic inclusion

$$x^* + \frac{g}{\Delta}(P - x^*) \subset L(g), \quad \frac{2rg}{\Delta} \mathbb{B}_{\mathcal{V}} \subset K_g. \quad (31)$$

The second inclusion follows by taking differences of the homothetic copy of the ball in (25). Together with $K_g \subset 2CE_F(x)$ it implies, for each Euclidean unit h , $\|(2rg/\Delta)h\|_x \leq 2C$. This is the upper bound in (28). Taking ratios proves (29), and (5) gives (30). $\square$

For geometric interpretation, the inradius r_g is the minimum width of $L(g)$, where width means the distance between parallel supporting hyperplanes. Indeed, the support function of K_g in a unit direction h is

$$\max_{u \in L(g)} \langle h, u \rangle - \min_{v \in L(g)} \langle h, v \rangle.$$

A centered ball is contained in a closed convex body exactly when its support function is no larger in every direction. Taking the minimum over h proves the assertion. Since $x \in L(g)$ has gap g and $\Phi \subset L(g)$, the objective-direction width is exactly $g/\|c_{\mathcal{V}}\|$. Thus

$$\frac{2rg}{\Delta} \leq r_g \leq \frac{g}{\|c_{\mathcal{V}}\|}. \quad (32)$$

Conditioning is therefore the squared aspect ratio of the sublevel difference body, within barrier-parameter factors; the matching $D(g)/g$ law results from the two bounds on its smallest width.

Corollary 3.4 (Uniformity and accuracy rates). *For every fixed barrier on the fixed compact instance,*

$$\hat{\kappa}_F(g) = \Theta((D(g)/g)^2) \quad (g \downarrow 0). \quad (33)$$

The same two-sided estimates hold uniformly for barriers that may depend on g , provided their parameters satisfy $\nu_F \leq \bar{\nu}$ and the points in question have residuals at most $\bar{\rho} < 1$. For exact centers, all such barriers attain every gap in

$$0 < g < \frac{\Delta}{1 + C_{\bar{\nu}}}. \quad (34)$$

If $D(g) = \Theta(g^\theta)$ with $0 \leq \theta \leq 1$, every fixed barrier has $\hat{\kappa}_F(g) = \Theta(g^{2\theta-2})$, and equivalently $\kappa_F(x_F(\mu)) = \Theta(\mu^{2\theta-2})$.

Proof. The first and the residual-uniform assertions follow directly from (30) with $C \leq C(\bar{\nu}, \bar{\rho})$. At the analytic center, $\nabla F = 0$, so lemma 3.1 with $\rho = 0$ contains all of P in $x_F^{\text{ac}} + C_{\nu_F} E_F(x_F^{\text{ac}})$. If y maximizes the objective on P , then (11) gives

$$\Delta - a_F = \langle c, y - x_F^{\text{ac}} \rangle \leq C_{\nu_F} \|c_{\mathcal{V}}\|_{x_F^{\text{ac}}}^* \leq C_{\nu_F} a_F.$$

Thus $a_F \geq \Delta/(1 + C_{\nu_F})$, proving (34). The exponent follows by substitution and proposition 2.3. $\square$

The constants are explicit and separate three sources of dependence: the barrier parameter, the allowed residual, and the fixed geometry $(r, \Delta, \|c_{\mathcal{V}}\|)$. Uniformity over a family of problem instances additionally requires control of these geometric constants. No uniform conclusion is asserted when the barrier parameters diverge or the residual bounds approach one. In particular, the law is an accuracy-rate statement, not a dimension-independent runtime estimate.

3.3 Canonical constants and compact-sublevel localization

For canonical barriers one can sometimes improve the upper constant using dual slack information. This is a specialization, not an additional assumption needed for the general law. Consider either $P = \{x : Ax = b, x \geq 0\}$ with $F = -\sum_i \log x_i$, or $P = \{X : AX = b, X_j \succeq 0 (j = 1, \dots, m)\}$ with $F = -\sum_j \log \det X_j$. Assume the respective strictly positive or positive-definite relative interior is nonempty, and use the Euclidean or product Frobenius metric. Let ν_{can} denote the number of scalar coordinates or the sum of matrix-block sizes. At a central point, the canonical dual slacks satisfy $s_i = \mu/x_i$ or $S_j = \mu X_j^{-1}$. Hence if, on a tail $0 < \mu \leq \bar{\mu}$, their largest coordinate or operator norm is at most M , then, with $g = g_{\text{can}}(\mu)$,

$$\lambda_{\max}(\hat{H}_{\text{can}}(g)) \leq M^2 \mu^{-2}, \quad \hat{\kappa}_{\text{can}}(g) \leq \left(\frac{M \nu_{\text{can}} \ell(\hat{x}_{\text{can}}(g))}{g} \right)^2. \quad (35)$$

For the LP this follows by bounding $\sum_i h_i^2/x_i^2$; for the SDP, bound $\sum_j \|X_j^{-1/2} U_j X_j^{-1/2}\|_F^2$ by $\mu^{-2} M^2 \sum_j \|U_j\|_F^2$. Use $g \leq \nu_{\text{can}} \mu$ and (27) to obtain the condition-number bound.

Such a finite M follows directly from any fixed strictly feasible point z or Z . Let $\sigma > 0$ be its smallest coordinate or its smallest block eigenvalue. Centrality supplies a dual multiplier y_{μ} with $A^* y_{\mu} + s_{\mu} = c$ (using S_{μ} in the matrix case), where A^* is the adjoint of the equality-constraint map. Canonical complementarity gives $\langle b, y_{\mu} \rangle = \langle c, x_{\mu} \rangle - \nu_{\text{can}} \mu$. Consequently,

$$\langle z, s_{\mu} \rangle = \langle c, z \rangle - \langle b, y_{\mu} \rangle \leq g(z) + \nu_{\text{can}} \bar{\mu}, \quad M = \frac{g(z) + \nu_{\text{can}} \bar{\mu}}{\sigma} \text{ is a valid bound,}$$

with traces and block notation understood for the SDP. Positive slacks make the coordinate or operator-norm bound immediate. Thus this specialized constant requires neither nondegeneracy nor strict complementarity. In particular, the canonical barrier and every other fixed barrier have the same conditioning rate; the equal-gap comparison (23) also gives the converse comparison of their full spectra.

Global boundedness is convenient for central-path existence and for choosing Δ . The pointwise theorem only needs bounded lower sublevels, which is useful for standard-form instances with unbounded feasible regions.

Proposition 3.5 (Localization to a compact sublevel). *Let P instead be closed and convex, with nonempty relative interior, and let the nonconstant linear objective have a finite attained optimum. Suppose $L(\bar{g})$ is compact for some $\bar{g} > 0$. Let F satisfy definition 2.1 on P° . There is a ball*

$z + r\mathbb{B}_{\mathcal{V}} \subset L(\bar{g})$ with $r > 0$. For every point $x \in P^\circ$ with $0 < g(x) \leq \bar{g}$ and $\rho_F(x; \mu) \leq \rho < 1$, all pointwise conclusions of [theorems 3.2](#) and [3.3](#) hold with Δ replaced by $\bar{g}$ in their upper bounds. The spectral-width bounds of [section 4](#) below hold as well. This assertion does not require existence of a global central path on the unbounded set.

Proof. Take $x^* \in \Phi$ and $z_0 \in P^\circ$. A sufficiently short positive segment from x^* towards z_0 gives $z \in P^\circ$ with $g(z) < \bar{g}$. A small relative ball about z lies in P and below the level $\bar{g}$. Dikin containment and semiboundedness are local line arguments and remain valid. The set $L(g(x))$ is compact, so diameters and chord maxima are attained. The proofs above now apply, using

$$x^* + \frac{g}{\bar{g}}(L(\bar{g}) - x^*) \subset L(g), \quad \frac{2rg}{\bar{g}}\mathbb{B}_{\mathcal{V}} \subset K_g$$

in place of [\(31\)](#). All later width proofs use only these containments and bounded objective-decreasing rays in $L(g)$. $\square$

4 Directional widths determine the full spectrum

The diameter law uses the longest and shortest scales of a sublevel set. Intermediate eigenvalues depend on how many independent directions have each scale. Radial widths express this information without selecting a barrier or a center.

For $0 < g \leq \Delta$ and a Euclidean unit vector $h \in \mathcal{V}$, define the radial function of the difference body by

$$q_g(h) = \max\{t \geq 0 : th \in K_g\}. \quad (36)$$

The set K_g is compact, convex, symmetric, and contains a neighborhood of zero. Thus q_g is positive, finite, even, and continuous on the unit sphere. For example, continuity follows by writing $q_g(h)$ as the reciprocal of the Minkowski functional of K_g , which is a norm equivalent to the Euclidean norm. For $j = 1, \dots, d$, put

$$w_j^-(g) = \inf_{\substack{U \subset \mathcal{V} \\ \text{codim } U = j-1}} \sup_{\substack{h \in U \\ \|h\|=1}} q_g(h), \quad (37)$$

$$w_j^+(g) = \sup_{\substack{S \subset \mathcal{V} \\ \text{dim } S = j}} \inf_{\substack{h \in S \\ \|h\|=1}} q_g(h). \quad (38)$$

Here and below U, S are linear subspaces. These two minimax profiles need not coincide for a general convex body.

Theorem 4.1 (Radial width bounds). *Under [theorem 3.2](#), list the eigenvalues of $H_F(x)$ in nondecreasing order. Then for $j = 1, \dots, d$,*

$$\frac{1}{(w_j^+(g))^2} \leq \lambda_j(H_F(x)) \leq \frac{4C^2}{(w_j^-(g))^2}. \quad (39)$$

Moreover, $w_j^+(g) \leq w_j^-(g) \leq 2Cw_j^+(g)$, $w_1^+(g) = w_1^-(g) = D(g)$, and $w_d^+(g) = w_d^-(g) = r_g$.

Proof. The radial form of [\(22\)](#) is

$$\frac{1}{\sqrt{\langle H_F(x)h, h \rangle}} \leq q_g(h) \leq \frac{2C}{\sqrt{\langle H_F(x)h, h \rangle}}, \quad \|h\| = 1.$$

Equivalently, for the Rayleigh quotient $R(h) = \langle H_F(x)h, h \rangle$,

$$q_g(h)^{-2} \leq R(h) \leq 4C^2 q_g(h)^{-2}. \quad (40)$$

The two Courant–Fischer formulas in increasing eigenvalue order are

$$\lambda_j = \sup_{\text{codim } U=j-1} \inf_{h \in U, \|h\|=1} R(h), \quad \lambda_j = \inf_{\text{dim } S=j} \sup_{h \in S, \|h\|=1} R(h).$$

Apply the lower estimate in the second formula and the upper estimate in the first. Positivity and uniform boundedness away from zero justify taking reciprocal squared infima and suprema. Explicitly,

$$\begin{aligned} \lambda_j &\geq \inf_{\text{dim } S=j} \sup_{h \in S, \|h\|=1} q_g(h)^{-2} = \left(\sup_{\text{dim } S=j} \inf_{h \in S, \|h\|=1} q_g(h) \right)^{-2}, \\ \lambda_j &\leq 4C^2 \sup_{\text{codim } U=j-1} \inf_{h \in U, \|h\|=1} q_g(h)^{-2} = 4C^2 \left(\inf_{\text{codim } U=j-1} \sup_{h \in U, \|h\|=1} q_g(h) \right)^{-2}. \end{aligned}$$

These are (39).

Every j -dimensional S intersects every codimension- $(j-1)$ subspace U in a nonzero direction. A unit vector in that intersection shows $\inf_{h \in S} q_g(h) \leq \sup_{h \in U} q_g(h)$; taking the indicated extrema proves $w_j^+ \leq w_j^-$. Combining the two bounds in (39) gives $w_j^- \leq 2Cw_j^+$. For $j=1$ both profiles are the maximum radial extent, namely $D(g)$. For $j=d$ both are the minimum radial extent, namely the centered inradius r_g . $\square$

For an ellipsoid the two minimax profiles agree with its decreasingly ordered principal semiaxes, and the theorem reduces to the familiar reciprocal square relation between semiaxes and Hessian eigenvalues. For a general sublevel difference body, (39) is the corresponding variational statement. It can distinguish several spectral scales even when a single condition number cannot.

4.1 Directed exits and the sharper pointwise constant

A center-dependent chord construction gives a second useful profile with the factor C^2 instead of $4C^2$. Fix $x \in P^\circ$. For an unoriented line $[h] \subset \mathcal{V}$, choose its Euclidean unit representative $\hat{h}$ so that $\langle c, \hat{h} \rangle < 0$ whenever the objective is nonconstant on that line. If the line is orthogonal to c_V , choose the sign with the longer positive exit time, breaking a tie arbitrarily. Define

$$\tau_x([h]) = \sup\{t \geq 0 : x + s\hat{h} \in P^\circ \text{ for } 0 \leq s < t\}. \quad (41)$$

The chosen ray remains in $L(g(x))$ until it exits, so its exit point is in $L(g(x))$, and $0 < \tau_x([h]) \leq \ell(x)$. This remains true under [proposition 3.5](#). The sign convention can cause discontinuity at objective-orthogonal lines, which is why the next definitions use infima and suprema. Put

$$v_j^-(x) = \inf_{\text{codim } U=j-1} \sup_{h \in U, \|h\|=1} \tau_x([h]), \quad (42)$$

$$v_j^+(x) = \sup_{\text{dim } S=j} \inf_{h \in S, \|h\|=1} \tau_x([h]). \quad (43)$$

Corollary 4.2 (Directed-exit spectral bounds). *Under [theorem 3.2](#),*

$$\frac{1}{(v_j^+(x))^2} \leq \lambda_j(H_F(x)) \leq \frac{C^2}{(v_j^-(x))^2}, \quad j = 1, \dots, d. \quad (44)$$

Also $v_j^+(x) \leq v_j^-(x) \leq Cv_j^+(x)$ and $v_1^+(x) = v_1^-(x) = \ell(x)$.

Proof. The boundary exit has local distance at least one by open Dikin containment, and at most C by (20). Consequently, for every unit h ,

$$\tau_x([h])^{-2} \leq \langle H_F(x)h, h \rangle \leq C^2 \tau_x([h])^{-2}.$$

All exit times are bounded below by the radius of a fixed relative ball about x and above by $\ell(x)$. The minimax argument in the proof of [theorem 4.1](#) therefore applies, without continuity, and also proves $v_j^+ \leq v_j^-$. Combining the two bounds in (44) gives $v_j^- \leq C v_j^+$. Finally, for any $y \in L(g(x))$ distinct from x , the ray along $y - x$ reaches at least as far as y . The selected sign agrees with that ray when the objective strictly decreases, and the longer-exit convention can only increase its length when the objective is constant. Thus $\sup_{[h]} \tau_x([h]) \geq \|y - x\|$. Maximizing over y and using the reverse inequality already noted proves $v_1^+ = v_1^- = \ell(x)$. $\square$

The radial profiles describe the sublevel geometry alone and hence support comparison across barriers. The directed profiles retain the location of a particular iterate and sharpen constants there. Neither formulation assumes that the optimal face is nondegenerate or that the spectrum has only two clusters.

5 Conditioning regimes from objective geometry

The diameter law separates two questions: how quickly objective sublevels shrink, and how their shape enters a barrier Hessian. Error bounds answer the first question. Their connection to weak sharp minima and conic degeneracy is classical ([Burke and Ferris, 1993](#); [Sturm, 2000](#); [Lourenço, 2021](#)). Here every matching diameter estimate yields a matching conditioning rate for every fixed self-concordant barrier.

Corollary 5.1 (Error bounds and attained rates). *Suppose, for some $M > 0$, $\theta \in (0, 1]$, and all sufficiently small positive gaps,*

$$\text{dist}(y, \Phi) \leq M g(y)^\theta. \quad (45)$$

Then

$$\max \left\{ \text{diam } \Phi, \frac{g}{\|c_V\|} \right\} \leq D(g) \leq \text{diam } \Phi + 2M g^\theta. \quad (46)$$

If Φ is a singleton, every fixed barrier consequently has $\widehat{\kappa}_F(g) = O(g^{2\theta-2})$. More generally, whenever $D(g) = \Theta(g^\theta)$ for $\theta \in [0, 1]$,

$$\widehat{\kappa}_F(g) = \Theta(g^{2\theta-2}). \quad (47)$$

All corresponding μ -rates follow by replacing g with μ for a fixed barrier. The gap statements also hold at points with uniformly bounded barrier parameters and residuals bounded by a fixed $\bar{\rho} < 1$.

Proof. Choose nearest points in compact Φ for two members of $L(g)$ and apply the triangle inequality. Points already in Φ have zero distance, so this proves the upper bound for small g . The inclusion $\Phi \subset L(g)$ proves its first lower bound. For every $0 < g < \Delta$ there is a point $y \in P$ with gap exactly g , by taking a segment from an optimum to a maximizer. Then $g \leq \|c_V\| \|y - x^*\|$ for any $x^* \in \Phi$, which proves the second lower bound. Apply [theorem 3.3](#) and [proposition 2.3](#) to finish. $\square$

An upper error bound alone gives an upper conditioning rate at a unique optimum. A matching lower rate requires sublevel chords that attain its exponent. This distinction matters for singularity-degree bounds, as the paired examples in [section 7](#) demonstrate.

Proposition 5.2 (Faces, vertices, and curved optima). *For every fixed barrier, the following statements hold.*

1. *If $\dim \Phi > 0$, then $D(g) = \Theta(1)$ and $\widehat{\kappa}_F(g) = \Theta(g^{-2})$.*
2. *If P is a polytope and $\Phi = \{x^*\}$, then $D(g) = \Theta(g)$ and $\widehat{\kappa}_F(g) = \Theta(1)$. No nondegeneracy assumption on the vertex is needed.*
3. *If $\Phi = \{x^*\}$ and $D(g) = \Theta(\sqrt{g})$, then $\widehat{\kappa}_F(g) = \Theta(g^{-1})$.*

Proof. A positive-dimensional optimal face has positive diameter, while $D(g) \leq \text{diam } P$. For the polytope assertion let $v_1, \dots, v_k$ be its vertices other than x^* and put $\delta = \min_i g(v_i) > 0$. Every point can be written $y = \alpha_0 x^* + \sum_i \alpha_i v_i$, with nonnegative coefficients summing to one. Thus

$$\|y - x^*\| \leq \text{diam}(P) \sum_i \alpha_i \leq \frac{\text{diam}(P)}{\delta} g(y).$$

This gives $D(g) = O(g)$; the lower bound in (46) gives the converse. All three conditioning claims now follow from (47). $\square$

These are useful regimes, not an exhaustive classification of convex sets. The diameter law applies even when $D(g)$ has no power-law asymptotic. For a fixed compact LP, however, the singleton and positive-dimensional-face alternatives are exhaustive: its asymptotic conditioning exponent is zero or two. An intermediate slope measured over a finite LP accuracy range does not establish a fractional asymptotic class.

The polytope argument is the elementary finite-vertex form of weak sharpness. It also shows why primal degeneracy alone cannot force accuracy-driven ill-conditioning of the equality-reduced primal Hessian. The bounded limit for the logarithmic barrier is made explicit in section 6. Boundedness here is instancewise; its constant need not be small or uniform over the data.

Example 5.3 (A curved singleton in a 2×2 spectrahedron). Minimize X_{22} over $X \succeq 0$, $\text{tr } X = 1$. Write

$$X = \begin{pmatrix} 1-u & v \\ v & u \end{pmatrix}, \quad 0 \leq u \leq 1, \quad v^2 \leq u(1-u).$$

The unique optimum is $\text{Diag}(1, 0)$. At $u \leq g$, the Frobenius distance to this optimum is $O(\sqrt{g})$. The two feasible points with $u = g$ and $v = \pm\sqrt{g(1-g)}$ are separated by $2\sqrt{2g(1-g)}$, so $D(g) = \Theta(\sqrt{g})$. Hence every fixed barrier has condition number $\Theta(g^{-1})$.

For $F = -\log \det X$, symmetry forces $v = 0$ on the central path. Centrality in u gives

$$\mu = \frac{g(1-g)}{1-2g}, \quad 0 < g < \frac{1}{2}.$$

The coordinate directions $\partial X / \partial u$ and $\partial X / \partial v$ are orthogonal and both have squared Frobenius norm two. The two reduced-Hessian eigenvalues are therefore

$$\frac{1}{g(1-g)}, \quad \frac{1}{2}(g^{-2} + (1-g)^{-2}).$$

They are in this increasing order for sufficiently small g , and $\widehat{\kappa}_{\log}(g) \sim (2g)^{-1}$.

5.1 A uniform plateau near an optimal edge

Consider the fixed simplex with a varying objective,

$$P = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1, x \geq 0\}, \quad c_\vartheta = (0, \vartheta, 1), \quad 0 < \vartheta \leq 1. \quad (48)$$

For each positive ϑ the unique optimum is e_1 . At $\vartheta = 0$ the optimal set becomes an edge. This family distinguishes the asymptotic accuracy rate from a large constant caused by a nearly optimal face.

Proposition 5.4 (The two gap scales of the simplex). *Uniformly for $0 < \vartheta \leq 1$ and $0 < g \leq 1$,*

$$D_\vartheta(g) = \Theta(\min\{g/\vartheta, 1\}). \quad (49)$$

For any family of barriers with parameters at most $\bar{\nu}$, on a common sufficiently small gap interval independent of ϑ ,

$$\widehat{\kappa}_{F,\vartheta}(g) = \Theta(\min\{\vartheta^{-2}, g^{-2}\}), \quad (50)$$

with constants depending only on $\bar{\nu}$. The same estimate holds at existing approximately central points with a common residual bound $\bar{\rho} < 1$, with constants also depending on $\bar{\rho}$.

Proof. For $x \in L_\vartheta(g)$, $\vartheta(x_2 + x_3) \leq \vartheta x_2 + x_3 \leq g$, while $x_2 + x_3 \leq 1$. Hence

$$\|x - e_1\| \leq \sqrt{2} \min\{g/\vartheta, 1\}.$$

The edge point $(1 - t, t, 0)$, $t = \min\{g/\vartheta, 1\}$, belongs to $L_\vartheta(g)$ and has distance $\sqrt{2}t$ from e_1 . Taking diameters proves (49) with absolute constants. The set P and its interior-ball radius are fixed, its objective range is $\Delta = 1$, and

$$\|(c_\vartheta)_\nu\|^2 = \frac{2}{3}(\vartheta^2 - \vartheta + 1) \in [1/2, 2/3].$$

The diameter law is therefore uniform in ϑ . The common attained gap interval follows from [corollary 3.4](#). $\square$

For the logarithmic barrier the limiting plateau height is explicit:

$$\lim_{\mu \downarrow 0} \kappa_{\log, \vartheta}(x_{\log, \vartheta}(\mu)) = \frac{1 + \vartheta^2 + \sqrt{1 - \vartheta^2 + \vartheta^4}}{1 + \vartheta^2 - \sqrt{1 - \vartheta^2 + \vartheta^4}}, \quad (51)$$

and multiplying this limit by ϑ^2 and then letting $\vartheta \downarrow 0$ gives $4/3$. Indeed, write the equality multiplier as $\alpha_\mu = \mu/x_1 > 0$. Centrality gives

$$x_2 = \frac{\mu}{\vartheta + \alpha_\mu}, \quad x_3 = \frac{\mu}{1 + \alpha_\mu}, \quad \alpha_\mu \rightarrow 0.$$

The last limit follows from $x(\mu) \rightarrow e_1$, itself implied by $g(\mu) \rightarrow 0$. On the tangent vector $h = (-u - v, u, v)$ the Rayleigh quotient of $\mu^2 H_{\log}$ converges to

$$\frac{\vartheta^2 u^2 + v^2}{2(u^2 + uv + v^2)}.$$

Its eigenvalues in the fixed Euclidean metric are $(1 + \vartheta^2 \pm \sqrt{1 - \vartheta^2 + \vartheta^4})/3$, which proves (51). Their product is $\vartheta^2/3$, and the larger tends to $2/3$ as $\vartheta \downarrow 0$, proving the stated $4/3$ limit. This is an iterated limit: μ first tends to zero with ϑ fixed. The uniform joint statement is (50), which describes the transition near $g \asymp \vartheta$ without interchanging limits.

6 The limiting Hessian at a unique LP optimum

For a unique polytope optimum the diameter law gives bounded conditioning for every fixed barrier. The logarithmic barrier has a stronger conclusion: a positive-definite scaled matrix limit, including at a degenerate vertex. We present it as a classical corollary of LP central-path endpoint theory, not as a new convergence theorem. Adler and Monteiro (1991, Theorem 3.3) identify the dual analytic-center limit; their Theorem 5.1 and subsequent discussion explicitly cover the full-column-rank active matrix at a unique optimum. Weighted path limits and analytic extension were further developed by Güler (1994); Halická (1999). The compression below makes the consequence for the equality-reduced Hessian explicit.

Let

$$P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}, \quad A \in \mathbb{R}^{m \times n}, \quad \text{rank } A = m < n, \quad (52)$$

be compact and contain a strictly positive point. Suppose $c^\top x$ has a unique optimum x^* . Use $F_{\log}(x) = -\sum_i \log x_i$ and an orthonormal basis matrix $W \in \mathbb{R}^{n \times (n-m)}$ of $\ker A$. The centrality equations supply positive slacks $s(\mu)$ and multipliers $y(\mu)$ with

$$Ax(\mu) = b, \quad A^\top y(\mu) + s(\mu) = c, \quad x_i(\mu)s_i(\mu) = \mu. \quad (53)$$

In particular strict dual feasibility holds. Define $B = \{i : x_i^* > 0\}$ and $N = \{1, \dots, n\} \setminus B$. Classical endpoint theory gives

$$s(\mu) \longrightarrow s^a, \quad s_B^a = 0, \quad s_N^a > 0. \quad (54)$$

Here s^a is the analytic center of the dual optimal slack face. The strict inequality on N is the strict complementarity of the centered LP optimum; it does not assert primal nondegeneracy.

Proposition 6.1 (A classical endpoint corollary). *For the data above,*

$$\mu^2 H_{\log}(x(\mu)) \longrightarrow L := W^\top \text{Diag}((s^a)^2)W \succ 0, \quad (55)$$

where matrices represent operators in the basis W . Therefore

$$\mu^2 \lambda_j(H_{\log}(x(\mu))) \longrightarrow \lambda_j(L) \quad (1 \leq j \leq n-m), \quad \kappa_{\log}(x(\mu)) \longrightarrow \kappa(L) < \infty. \quad (56)$$

The conclusion holds when $|B| < m$, the degenerate-vertex case.

Proof. Complementarity yields the exact identity

$$\mu^2 W^\top \text{Diag}(x(\mu)^{-2})W = W^\top \text{Diag}(s(\mu)^2)W.$$

The right side converges in operator norm by (54). To prove that its limit is positive definite, first observe that A_B has full column rank. Otherwise a nonzero $v \in \ker A_B$ gives feasible points $(x_B^* \pm tv, 0_N)$ for sufficiently small $t > 0$. Since $c_B = A_B^\top y^a$ by complementary slackness, both points have the optimal objective value, contradicting uniqueness.

If $z \in \ker A$ has $z^\top \text{Diag}((s^a)^2)z = 0$, positivity of s_N^a gives $z_N = 0$. Then $A_B z_B = 0$ and full column rank imply $z = 0$. Hence the quadratic form is positive definite on $\ker A$. Continuity of ordered eigenvalues and of the condition number on positive-definite matrices proves (56). $\square$

6.1 An active-set formula and quantitative constants

Write a_i for column i of A . The limiting slack can be computed from the active set without following the central path. It is $s^a = c - A^\top y^a$, where

$$\begin{aligned} y^a &= \arg \max_y \sum_{i \in N} \log(c_i - a_i^\top y) \\ \text{subject to } & A_B^\top y = c_B, \quad c_N - A_N^\top y > 0. \end{aligned} \quad (57)$$

Every feasible multiplier is optimal because $b = A_B x_B^*$. For clarity, existence and uniqueness of this maximizer also follow directly once strict complementarity is known. If $z > 0$ is a fixed primal feasible point and s is dual optimal, then $z^\top s = c^\top z - \text{OPT}$ bounds every coordinate of s . Full row rank makes $y \mapsto A^\top y$ injective, so the closed dual optimal face is bounded. The logarithmic objective diverges to $-\infty$ where an N -slack vanishes. On the tangent space $\ker A_B^\top$, its negative Hessian is

$$A_N \text{Diag}(s_N^{-2}) A_N^\top.$$

It is positive definite on that tangent space: if both $A_N^\top h$ and $A_B^\top h$ vanish, full row rank gives $h = 0$. Thus the maximizer is unique, with the zero-dimensional case interpreted as a single feasible multiplier.

Let R_N restrict a vector to its N -coordinates, and define

$$\sigma_N = \sigma_{\min}(R_N W) > 0, \quad \eta_N = \|R_N W\|_2 \leq 1, \quad s_{\min} = \min_{i \in N} s_i^a, \quad s_{\max} = \max_{i \in N} s_i^a.$$

The same uniqueness argument proves injectivity of $R_N W$. The bounds

$$s_{\min}^2 (R_N W)^\top (R_N W) \preceq L \preceq s_{\max}^2 (R_N W)^\top (R_N W) \quad (58)$$

give

$$s_{\min}^2 \sigma_N^2 \leq \lambda_{\min}(L) \leq s_{\max}^2 \sigma_N^2, \quad (59)$$

$$s_{\min}^2 \eta_N^2 \leq \lambda_{\max}(L) \leq s_{\max}^2 \eta_N^2, \quad (60)$$

$$\left(\frac{s_{\min}}{s_{\max}} \right)^2 \left(\frac{\eta_N}{\sigma_N} \right)^2 \leq \kappa(L) \leq \left(\frac{s_{\max}}{s_{\min}} \right)^2 \left(\frac{\eta_N}{\sigma_N} \right)^2. \quad (61)$$

These constants separate reduced-cost spread from the angle of $\ker A$ to the coordinate subspace supported on B . Neither factor is controlled by the statement that the optimum is unique. The simplex in [proposition 5.4](#) already shows unbounded limiting condition numbers within compact instances of fixed dimension.

Example 6.2 (A compact degenerate vertex). Take

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3/2 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 3/2 \end{pmatrix}, \quad c = (1, 1, 0, 1)^\top. \quad (62)$$

The simplex equality makes P compact. The point $(2, 3, 3, 3)^\top / 11$ is strictly feasible, and $y = (-1, 0)^\top$ is strictly dual feasible. The unique optimum is e_3 , so $|B| = 1 < m = 2$. Every dual optimum is $y = (-3t/2, t)^\top$, with

$$(s_1, s_2, s_4) = (1 + 3t/2, 1 + t/2, 1 - 3t/2), \quad -2/3 < t < 2/3$$

in its relative interior. Differentiating the sum of the three logarithms gives $27t^2 + 36t - 4 = 0$. Only one root lies in the displayed interval:

$$t_a = \frac{-6 + 4\sqrt{3}}{9}. \quad (63)$$

Strict concavity makes it the dual analytic center. For an exact calculation without choosing an orthonormal nullspace basis, write

$$z = Tu, \quad T = \begin{pmatrix} -1/3 & 1 \\ 1 & 0 \\ -2/3 & -2 \\ 0 & 1 \end{pmatrix}, \quad T^\top T = \begin{pmatrix} 14/9 & 1 \\ 1 & 6 \end{pmatrix}.$$

The columns of T span $\ker A$. The eigenvalues of L are the generalized eigenvalues of

$$T^\top \text{Diag}((s^a)^2)T = \begin{pmatrix} (s_1^a)^2/9 + (s_2^a)^2 & -(s_1^a)^2/3 \\ -(s_1^a)^2/3 & (s_1^a)^2 + (s_4^a)^2 \end{pmatrix}$$

against $T^\top T$. Numerically these are approximately 0.24888160 and 1.14289062, giving $\kappa(L) \approx 4.59210571$. The exact matrix formulas establish the finite limit; the decimals only identify its size.

The same endpoint proof applies to a strictly primal–dual feasible LP with an unbounded feasible region, whenever the classical central path has the stated centered limits. For example, for $M \geq 1$,

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad c = (0, 1, 1, M)^\top,$$

the unique optimum e_1 is degenerate and $x(\mu) = (1, \mu, \mu, \mu/M)^\top$, $s(\mu) = (\mu, 1, 1, M)^\top$ solve the centrality equations exactly. In the orthonormal tangent basis $(0, 1, 1, 0)^\top/\sqrt{2}$, $(0, 0, 0, 1)^\top$, one has $\mu^2 H_{\log} = \text{Diag}(1, M^2)$. Every positive objective sublevel is compact, so [proposition 3.5](#) also applies on its central tail. This example shows directly that even at a degenerate vertex the finite limit can be arbitrarily ill-conditioned.

7 Fractional spectra and the limits of singularity degree

Semidefinite sublevels need not have either of the polyhedral diameter rates. We analyze a trace-normalized version of the 3×3 instance in [Sturm \(2000, Example 2\)](#). The associated nested constraints and quarter-power error geometry are classical; see also [Drusvyatskiy and Wolkowicz \(2017, Theorem 4.5.1, Example 4.5.2, and Section 4.7\)](#). Our purpose is to compute its entire reduced log-barrier spectrum, transfer the resulting scales to every fixed barrier, and compare it with a section having the same singularity degree but a different conditioning exponent.

7.1 Two spectrahedra and their sharp diameters

Consider

$$\min X_{22} \quad \text{subject to} \quad X \in \mathbb{S}_+^3, \quad \text{tr } X = 1, \quad X_{33} = X_{12}, \quad (64)$$

and the restricted problem obtained by adding

$$X_{13} = X_{23} = 0. \quad (65)$$

Here $\mathbb{S}_+^3$ is the cone of real symmetric positive semidefinite matrices. Write their feasible sets as P and P_0 , and their sublevel diameters as D and D_0 , respectively, each in its inherited Frobenius metric.

Proposition 7.1 (Equal singularity degree, different conditioning). *Both problems have compact feasible sets with positive-definite feasible points and unique optimum $X^* = e_1 e_1^\top$. The optimality systems of both problems have singularity degree two. Nevertheless,*

$$D(g) = \Theta(g^{1/4}), \quad D_0(g) = \Theta(g^{1/2}). \quad (66)$$

Thus every fixed self-concordant barrier on the respective set satisfies

$$\widehat{\kappa}_F(g) = \Theta(g^{-3/2}) \quad \text{on } P, \quad \widehat{\kappa}_{F_0}(g) = \Theta(g^{-1}) \quad \text{on } P_0. \quad (67)$$

For the displayed conic formulations, neither problem has a strictly complementary optimal primal-dual pair.

We use the standard definition of singularity degree for a feasible conic *system*: it is the minimum number of facial-reduction steps required to reach a face in whose relative interior the affine system has a feasible point (Drusvyatskiy and Wolkowicz, 2017; Waki and Muramatsu, 2013). At each step an equality combination nonnegative on the current cone and zero on the affine system exposes a smaller face. In the proposition the system includes the optimality equality $X_{22} = 0$. The original feasible systems have singularity degree zero, because they contain positive-definite matrices. Keeping these systems distinct is essential when applying error bounds to objective accuracy.

Proof. Trace normalization gives compactness. The matrix

$$\begin{pmatrix} 2/5 & 1/5 & 0 \\ 1/5 & 2/5 & 0 \\ 0 & 0 & 1/5 \end{pmatrix}$$

is positive definite and feasible for both problems. Write a point of P as

$$X = \begin{pmatrix} a & b & t \\ b & u & e \\ t & e & b \end{pmatrix}, \quad a + u + b = 1. \quad (68)$$

If $u = 0$, the second row and column of a PSD matrix vanish, so $b = e = 0$. The third diagonal is then zero, so $t = 0$. Thus the unique optimum in both sets is X^* , with value zero.

For $X \in L(g)$, principal minors give

$$0 \leq b \leq \sqrt{g}, \quad t^2 \leq ab \leq \sqrt{g}, \quad e^2 \leq ub \leq g^{3/2}, \quad |a - 1| = u + b \leq g + \sqrt{g}. \quad (69)$$

Together with $0 \leq u \leq g$, these inequalities prove $\|X - X^*\|_F = O(g^{1/4})$. If $t = e = 0$, they instead give $\|X - X^*\|_F = O(g^{1/2})$. These prove both diameter upper bounds.

For $0 < g \leq 1/16$, set

$$u = g, \quad b = \frac{\sqrt{g}}{2}, \quad a = 1 - g - \frac{\sqrt{g}}{2}, \quad e = 0, \quad t = \pm \frac{g^{1/4}}{2\sqrt{2}}. \quad (70)$$

Here $a \geq 13/16$ and $ag - b^2 = g(a - 1/4) > 0$, while

$$\det X = agb - b^3 - gt^2 = g^{3/2}(a/2 - 1/4) > 0.$$

The leading principal minors are positive, so both matrices are positive definite. Their Frobenius distance is $g^{1/4}$, proving the first lower bound. For P_0 , take the same a, b, u and $t = e = 0$. The

resulting matrix is positive definite and its distance from X^* is at least $\sqrt{2}b = \sqrt{g/2}$, proving the second.

For the facial-reduction claim let $E_{ii} = e_i e_i^\top$ and $J_{ij} = (e_i e_j^\top + e_j e_i^\top)/2$, so that $\langle J_{ij}, X \rangle = X_{ij}$. The first equality $X_{22} = 0$ exposes the face supported on $\text{span}\{e_1, e_3\}$. On this face $X_{12} = 0$, so $X_{33} - X_{12} = 0$ exposes the ray generated by E_{11} . Its relative interior contains X^* , proving that two steps suffice in both systems.

To exclude one step, an exposing combination for either optimality system has the form

$$Z = \alpha I + \beta(E_{33} - J_{12}) + \gamma E_{22} + \delta J_{13} + \eta J_{23} \succeq 0, \quad (71)$$

where $\delta = \eta = 0$ for P . Its value on the affine system must be zero; the trace equation is the only equation with nonzero right side, so $\alpha = 0$. Equivalently $Z_{11} = 0$. A PSD matrix with zero diagonal entry has a zero corresponding row. The first row therefore forces $\beta = \delta = 0$; then $Z_{33} = 0$ forces $\eta = 0$. Consequently every nonzero first-step certificate is a positive multiple of E_{22} . It cannot expose the final ray. This proves minimality of the two-step chain.

The same calculation establishes the complementarity assertion. A dual slack has the form (71) with $\gamma = 1$, since the objective is X_{22} ; optimality makes $\alpha = 0$. The only optimal slack is therefore E_{22} . It has rank one, as does X^* , so their ranks sum to two rather than three. Dual strict feasibility nonetheless holds: choosing a negative multiplier for the trace equation and all other multipliers zero makes the slack $E_{22} + \tau I \succ 0$ for $\tau > 0$. Finally, [theorem 3.3](#) proves (67) for every fixed barrier. $\square$

For a singleton SDP optimum, the classical singularity-degree error bound yields $D(g) = O(g^{2-s})$ when the optimality system has degree s , on a fixed compact set ([Sturm, 2000](#); [Drusvyatskiy and Wolkowicz, 2017](#)). Indeed, a feasible point has zero cone residual and violates only the added optimality equation by g ; its distance to the optimal affine space is at most a fixed multiple of that residual, by a right inverse on the range of the equality map. The error bound then controls its distance to the optimum. If this exponent is attained, the diameter law gives

$$\hat{\kappa}_F(g) = \Theta(g^{2^{1-s}-2}). \quad (72)$$

The two degree-two systems above show why attainment must be verified: the first has the quarter-power diameter, whereas the second has a strictly faster square-root decay. Thus singularity degree alone does not determine the conditioning exponent, even for these compact 3×3 formulations. The diameter records sublevel shrinkage that this facial-reduction count does not determine.

7.2 The exact log-barrier center and four spectral scales

The general diameter theorem gives the condition-number exponent on P . In this example one can also compute every eigenvalue scale. This supplements the geometric argument with an exact spectral description, including its leading constants.

Theorem 7.2 (Four reduced-Hessian scales). *For $F(X) = -\log \det X$ on (64), the central point at gap g has the form*

$$\hat{X}_{\log}(g) = \begin{pmatrix} a & b & 0 \\ b & g & 0 \\ 0 & 0 & b \end{pmatrix}, \quad b = \frac{-g + \sqrt{3g - 2g^2}}{3}, \quad a = 1 - g - b, \quad (73)$$

and the corresponding central-path parameter satisfies

$$q = ag - b^2, \quad \mu = \frac{q}{1 - b - 2g}. \quad (74)$$

These formulas apply for all sufficiently small positive g . For the four increasingly ordered eigenvalues of the reduced Hessian in the Frobenius metric, as $\mu \downarrow 0$,

$$\lambda_1 \sim \sqrt{2}\mu^{-1/2}, \quad \lambda_2 \sim \mu^{-1}, \quad \lambda_3 \sim \sqrt{2}\mu^{-3/2}, \quad \lambda_4 \sim \frac{4}{7}\mu^{-2}. \quad (75)$$

In particular,

$$\kappa_{\log}(X_{\log}(\mu)) \sim \frac{2\sqrt{2}}{7}\mu^{-3/2} \sim \frac{3\sqrt{3}}{7}g^{-3/2}. \quad (76)$$

On P_0 , the log-barrier center is the same matrix. Its two reduced-Hessian eigenvalues are asymptotic to μ^{-1} and $(4/7)\mu^{-2}$, respectively, so its condition number is asymptotic to $(4/7)\mu^{-1}$, or $(6/7)g^{-1}$.

Proof. Conjugation by $\text{Diag}(1, 1, -1)$ preserves the feasible set, the objective, and F . Uniqueness of the central point therefore forces $t = e = 0$ in (68). At a fixed gap $u = g$, the remaining free coordinate is b , and

$$F = -\log b - \log q, \quad q = g(1 - g - b) - b^2.$$

Stationarity in b gives $q + bq_b = 0$, or

$$3b^2 + 2gb + g^2 - g = 0.$$

The positive root is the value in (73). Stationarity of $g/\mu + F$ in g gives $1/\mu = q_g/q$, which is (74). For small $g > 0$, $b > 0$, $a > 0$, $q > 0$, and $q_g > 0$. Thus these matrices are interior stationary points for a positive μ , hence are the unique central points by strict convexity. Conversely uniqueness and the fixed-gap stationarity determine every small-gap center. The same argument holds on P_0 , where $t = e = 0$ are imposed as constraints.

For an explicit parametrization of the entire small-parameter tail, put $\tau = g/b$ and $d = \tau^2 + 2\tau + 3$. The stationarity equations become

$$a = \frac{\tau + 3}{d}, \quad b = \frac{\tau}{d}, \quad g = \frac{\tau^2}{d}, \quad q = \frac{\tau^2(\tau + 2)}{d^2}, \quad \mu = \frac{\tau^2(\tau + 2)}{d(3 + \tau - \tau^2)}. \quad (77)$$

For $0 < \tau < 1$, all required positivity conditions hold. Both g and μ extend continuously to $[0, 1]$, taking values 0 and $1/6$ at the endpoints. Thus every gap and every central parameter in $(0, 1/6)$ is attained. Moreover $\tau^2 \leq 6g$ and $\tau^2 \leq 12\mu$ on this interval, so any such parameter choices tend to zero as g or μ tends to zero. Consequently the limits obtained from (77) describe the entire central path tail, rather than a selected sequence of central points.

As $g \downarrow 0$, the formulas give

$$b \sim \sqrt{g/3}, \quad q \sim 2g/3, \quad \mu \sim 2g/3; \quad a \rightarrow 1, \quad b \sim \sqrt{\mu/2}, \quad q \sim \mu. \quad (78)$$

We now compute the Hessian in the four tangent coordinates (b, g, t, e) . The (b, g) and (t, e) subspaces are orthogonal in the Frobenius metric, and the Hessian has no mixed terms between them at $t = e = 0$, either by the same sign symmetry or by direct differentiation.

Let $A_2 = \begin{pmatrix} a & b \\ b & g \end{pmatrix}$, with determinant q . For an off-block tangent perturbation $U = \begin{pmatrix} 0_{2 \times 2} & v \\ v^\top & 0 \end{pmatrix}$, where $v = (t, e)^\top$, one has

$$D^2F(X)[U, U] = \text{tr}(X^{-1}UX^{-1}U) = \frac{2}{b}v^\top A_2^{-1}v, \quad \|U\|_F^2 = 2v^\top v.$$

Thus this block's eigenvalues are those of A_2^{-1}/b . The larger eigenvalue of A_2 tends to one, and the smaller is asymptotic to $q \sim \mu$, by their product. This gives precisely $\sqrt{2}\mu^{-1/2}$ and $\sqrt{2}\mu^{-3/2}$.

On the (b, g) subspace, the tangent Gram matrix and the coordinate Hessian K are

$$G = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \quad K = \begin{pmatrix} b^{-2} + q_b^2/q^2 + 2/q & q_b q_g/q^2 + 1/q \\ q_b q_g/q^2 + 1/q & q_g^2/q^2 + 2/q \end{pmatrix}, \quad (79)$$

where $q_b = -g - 2b$ and $q_g = 1 - b - 2g$. In particular the coordinate Hessian K alone does not give the Euclidean operator's eigenvalues; they solve $Kv = \lambda Gv$. From (78),

$$\mu K_{bb} \rightarrow 6, \quad \mu^{3/2} K_{bg} \rightarrow -\sqrt{2}, \quad \mu^2 K_{gg} \rightarrow 1. \quad (80)$$

It follows that $\mu^2 G^{-1/2} K G^{-1/2}$ converges to the rank-one matrix $G^{-1/2} e_2 e_2^\top G^{-1/2}$, whose nonzero eigenvalue is $(G^{-1})_{22} = 4/7$. Hence the larger generalized eigenvalue is asymptotic to $(4/7)\mu^{-2}$. Also

$$\mu^3 \det K \rightarrow 6 - 2 = 4, \quad \det G = 7.$$

The product of the generalized eigenvalues is $\det K / \det G$, so the smaller is asymptotic to μ^{-1} . The four distinct powers determine their eventual increasing order, proving (75). Their extreme ratio proves (76), using $g \sim 3\mu/2$. Restricting to P_0 retains precisely the (b, g) block and gives its two stated eigenvalues and ratio. $\square$

The log-barrier calculation in [theorem 7.2](#) has a Lean formalization in the accompanying repository. It proves that the matrix is the unique minimizer of the actual barrier objective over all positive-definite feasible matrices, computes the second Fréchet derivative of $-\log \det$, identifies its operator in the Frobenius metric, and factors its characteristic polynomial. The formalization also proves the ordered asymptotics and both condition-number constants, including the restricted problem and coverage of the full small-parameter tail. The verification report is `formal/FRACTIONAL_SDP.md`. Its scope is the log-barrier spectrum; the singularity-degree and diameter assertions in [proposition 7.1](#), and the following transfer to other barriers, are separate mathematical results outside this formalization.

At equal gap the Loewner comparison in [theorem 3.2](#) transfers more than the extreme ratio. Every fixed barrier on P has ordered eigenvalues of orders $g^{-1/2}$, g^{-1} , $g^{-3/2}$, and g^{-2} in the same Frobenius metric. Every fixed barrier on P_0 has orders g^{-1} and g^{-2} . The leading constants in (75) concern the canonical log barrier; they are not asserted to be invariant under barrier changes.

SDP spectral clustering has substantial precedent. In particular, [Alizadeh et al. \(1998, Section 4\)](#) analyze eigenvalue behavior of a primal–dual Schur matrix, and [Peña \(2002, Section 3.3\)](#) compares central-path system conditioning. Those results concern a different matrix and, in the cited SDP clustering setting, strict complementarity. [Wei and Wolkowicz \(2010\)](#) construct instances with prescribed failure of strict complementarity and investigate their numerical difficulty.

[Sremac et al. \(2021, Section 4\)](#) relate singularity degree to spectral rates along an external regularization path: $X(\alpha)$ maximizes $\log \det X$ subject to $A(X) = b + \alpha A(B)$, with fixed $B \succ 0$, and $Z(\alpha) = \alpha X(\alpha)^{-1}$. Under their assumptions, Corollary 4.3 separates eigenvalue rates of $X(\alpha)$ and $Z(\alpha)$, Theorem 4.4 characterizes degree one through linear decay of the vanishing primal eigenvalues, and Theorem 4.7 gives separated rates for dual diagonal blocks. These are relevant antecedents for the degree–spectrum connection. Here the affine constraints remain fixed, the parameter is the primal objective gap, and the operator is the equality-reduced Hessian. Primal-iterate eigenvalues alone do not determine its compression to the equality tangent space.

The calculation above concerns the equality-reduced primal log-det Hessian at a non-strictly-complementary optimum, with its external metric specified. To the best of our knowledge, the four leading constants in (75) have not previously been stated for this trace-normalized classical construction. The novelty claimed here is this explicit spectral calculation and its all-barrier transfer, not the construction of a fractional SDP error bound or the general study of SDP spectral clusters.

8 LP spectral scales and the Newton right-hand side

The diameter law determines the ratio of the spectral edges. For a polytope one can say more: the dimension of the optimal face determines the number of eigenvalues at each of two scales. The logarithmic-barrier splitting is classical active-set geometry, closely related to the Hessian analyses of [Wright \(1994\)](#); [Forsgren et al. \(2002\)](#). Our use of [theorem 3.2](#) transfers every ordered scale to arbitrary fixed finite-parameter barriers. The eigenspaces need not coincide, and their different rates of rotation matter for the actual Newton equation.

8.1 The optimal-face tangent and the two spectral scales

Use the compact, strictly feasible standard-form representation (52), now allowing an optimal face Φ of any dimension f . Let $d = n - m$, and identify $\mathcal{V} = \ker A$ with $\mathbb{R}^d$ using the orthonormal matrix W . Set

$$B = \{i : x_i > 0 \text{ for some } x \in \Phi\}, \quad N = \{1, \dots, n\} \setminus B, \quad \mathcal{T} = \{z \in \ker A : z_N = 0\}.$$

Then $\mathcal{T}$ is exactly the tangent space of $\text{aff } \Phi$, and $\dim \mathcal{T} = f$. Indeed, a relative-interior point x^a of Φ is positive on B , so both signs of every sufficiently short displacement supported on B and satisfying $Az = 0$ stay feasible. Optimality then forces its objective increment to vanish, so both points lie in Φ . Conversely every chord of Φ vanishes on N . The objective annihilates this tangent: $c_{\mathcal{V}} \perp \mathcal{T}$. Since the objective is nonconstant, $f < d$. Write $P_{\mathcal{T}}$ for its orthogonal projector.

Classical logarithmic central-path endpoint theory ([Adler and Monteiro, 1991](#); [Güler, 1994](#); [Halická, 1999](#)) gives

$$x_B(\mu) \longrightarrow x_B^a > 0, \quad s_N(\mu) \longrightarrow s_N^a > 0, \quad (81)$$

where the limiting pair is strictly complementary, even when the LP is degenerate. Positivity is componentwise. The reduced logarithmic Hessian has the exact decomposition

$$\begin{aligned} H_{\log}(x(\mu)) &= H_N(\mu) + H_B(\mu), \\ H_N &= \mu^{-2} W^{\top} R_N^{\top} \text{Diag}(s_N^2) R_N W, \\ H_B &= W^{\top} R_B^{\top} \text{Diag}(x_B^{-2}) R_B W, \end{aligned} \quad (82)$$

with coordinate restrictions R_B, R_N . Thus $\ker H_N = \mathcal{T}$ under the stated identification, while $\|H_B\| = O(1)$.

Theorem 8.1 (Two scales for every fixed barrier). *Let F be any fixed ν -self-concordant barrier on P° . In increasing eigenvalue order, as $g \downarrow 0$,*

$$\lambda_j(\widehat{H}_F(g)) = \begin{cases} \Theta(1), & 1 \leq j \leq f, \\ \Theta(g^{-2}), & f < j \leq d. \end{cases} \quad (83)$$

The first case is empty when the optimum is unique. All constants may depend on the fixed instance and ν . For sufficiently small g , let $\Pi_F(g)$ be the orthogonal projector onto the lowest f eigenvalues; put $\Pi_F = 0$ if $f = 0$. Then

$$\|\Pi_F(g) - P_{\mathcal{T}}\| = O(g), \quad (84)$$

$$\|\Pi_{\log}(g) - P_{\mathcal{T}}\| = O(g^2). \quad (85)$$

In particular, the weak spectral subspace has the fixed face tangent as its limit, although it need not equal that tangent at positive gap.

Proof. The nonzero singular values of the fixed restriction $R_N W$ are bounded away from zero. By (81), the restriction of H_N to $\mathcal{T}^\perp$ has all its eigenvalues between $a\mu^{-2}$ and $b\mu^{-2}$, for fixed $a, b > 0$ on a central tail. Also $0 \preceq H_B \preceq b'I$. Weyl's inequalities give $d - f$ eigenvalues of order μ^{-2} . The minimax principle applied to $\mathcal{T}$ bounds the lowest f eigenvalues by b' . Dikin containment in the compact set P gives the common lower bound $H_{\log} \succeq (\text{diam } P)^{-2}I$. Therefore those f eigenvalues are of order one. By (15), $g = \Theta(\mu)$.

At equal objective gaps, (23) gives constants $a_F, b_F > 0$ such that

$$a_F \hat{H}_{\log}(g) \preceq \hat{H}_F(g) \preceq b_F \hat{H}_{\log}(g). \quad (86)$$

The eigenvalue minimax principle proves (83). Moreover, for another constant $a' > 0$,

$$\hat{H}_F(g) \succeq a' g^{-2} P_{\mathcal{T}^\perp}. \quad (87)$$

If Q_F is an orthonormal basis for the weak eigenspace, its compressed Hessian has norm $O(1)$. Evaluating (87) on $Q_F u$ gives $\|P_{\mathcal{T}^\perp} Q_F u\| \leq O(g) \|u\|$.

For the logarithmic barrier the eigenvector equation gives a sharper estimate. Write $H_{\log} Q_{\log} = Q_{\log} \Lambda_w$, with $\|\Lambda_w\| = O(1)$. Then

$$H_N Q_{\log} = Q_{\log} \Lambda_w - H_B Q_{\log}, \quad \|H_N Q_{\log}\| = O(1).$$

The inverse of H_N on $\mathcal{T}^\perp$ has norm $O(g^2)$, so $\|P_{\mathcal{T}^\perp} Q_{\log}\| = O(g^2)$. For subspaces of the same dimension, the norm of the difference of their orthogonal projectors equals the largest sine of their principal angles, here $\|P_{\mathcal{T}^\perp} Q_F\|$. This identity follows by taking a singular value decomposition of the matrix of inner products of their orthonormal bases. It proves both asserted projector rates. This is the elementary spectral-subspace perturbation argument underlying the familiar Davis–Kahan estimates (Davis and Kahan, 1970). $\square$

The theorem uses endpoint convergence only for the reference logarithmic barrier. It assumes no endpoint limit for the general barrier. Its constants are uniform over barriers with uniformly bounded ν , by theorem 3.2. The ordered eigenvalue scales and the general $O(g)$ projector estimate also hold at points satisfying (17) with a uniform bound $\rho < 1$, using their observed objective gap. The right-hand-side statement below has the additional requirement of exact centrality.

8.2 Exact Newton forcing and the meaning of weak-mode suppression

At an exact center $x_F(\mu_0)$, changing the central-path parameter to μ_1 gives the Newton equation for $F(x) + \langle c, x \rangle / \mu_1$:

$$H_F(x_F(\mu_0)) \Delta x = -(\mu_1^{-1} - \mu_0^{-1}) c_V. \quad (88)$$

The tangent derivative of the path has the same forcing direction, since $H_F(x_F(\mu)) x'_F(\mu) = \mu^{-2} c_V$. Thus, for every nonzero right-hand side r in (88),

$$\frac{\|\Pi_F(g)r\|}{\|r\|} = O(g), \quad \frac{\|\Pi_{\log}(g)r\|}{\|r\|} = O(g^2). \quad (89)$$

These follow immediately from $P_{\mathcal{T}} c_V = 0$ and theorem 8.1. The quantities in (89) are component *norm* fractions; the corresponding squared fractions are $O(g^2)$ and $O(g^4)$. At an approximately central point the Newton right-hand side has an additional centering residual. The same estimates for that actual right-hand side do not follow from spectral clustering alone.

For any positive-definite H and any orthogonal spectral projector Π , discard its subspace and set

$$u_s = H^{-1}(I - \Pi)r, \quad r - Hu_s = \Pi r, \quad H^{-1}r - u_s = H^{-1}\Pi r. \quad (90)$$

The first identity gives a relative Euclidean residual exactly equal to $\|\Pi r\| / \|r\|$. It does not bound the relative solution error by that same fraction. Conditional on having the spectral projection and a representation of the restricted operator, solving on the strong LP subspace has a bounded internal condition number. If its computed solution satisfies

$$\tilde{u}_s \in \text{range}(I - \Pi), \quad \|(I - \Pi)r - H\tilde{u}_s\| \leq \delta \|(I - \Pi)r\|,$$

then orthogonality gives

$$\frac{\|r - H\tilde{u}_s\|^2}{\|r\|^2} \leq \frac{\|\Pi r\|^2}{\|r\|^2} + \delta^2. \quad (91)$$

Constructing the projection and meeting a path-following algorithm's own error criterion are additional tasks. A Euclidean residual bound alone proves neither accurate Newton directions nor convergence of an inexact interior-point method.

8.3 A sharp general-barrier example

The rectangle below also has the strictly feasible standard-form embedding $(x, y) \mapsto (1 + x, 1 - x, y, 1 - y)$. This embedding multiplies the tangent metric by the scalar two, so it preserves eigenvector angles and all gap exponents. Thus the example tests the preceding LP rate directly.

Oscillatory perturbations within the self-concordant class have prior examples: [Duistermaat \(2001, Remark 2.3\)](#) uses $f + A \sin f$ to produce rescaled derivatives without boundary limits. The mixed perturbation below is used for the specific sharp eigenspace rate and residual–solution comparison, rather than as a first example of oscillatory barrier derivatives.

The following finite-parameter barrier shows that the $O(g)$ norm rate cannot in general be improved to $O(g^2)$. It also makes the distinction in (90) concrete.

Proposition 8.2 (Sharp rotation and small residual with large direction error). *On $P = [-1, 1] \times [0, 1]$, minimize y and set*

$$\begin{aligned} F(x, y) &= 4(F_0(x, y) + \varepsilon\psi(x, y)), \\ F_0(x, y) &= -\log(1 - x^2) - \log y - \log(1 - y), \\ \psi(x, y) &= x \sin(\log y), \quad \varepsilon = 1/100. \end{aligned} \quad (92)$$

This is a 20-self-concordant barrier. At gaps $g_k = \exp(-2\pi k)$, $k = 1, 2, \dots$, its exact center is $(0, g_k)$. For its weak spectral projector and $r = (0, 1)^\top$,

$$\|\Pi_F(g_k)r\| \sim \varepsilon g_k. \quad (93)$$

With $H = \widehat{H}_F(g_k)$ and $u_s = H^{-1}(I - \Pi_F(g_k))r$,

$$\frac{\|r - Hu_s\|}{\|r\|} \sim \varepsilon g_k, \quad \frac{\|H^{-1}r - u_s\|}{\|H^{-1}r\|} \longrightarrow 1. \quad (94)$$

Proof. We first check the barrier assumptions on the entire open rectangle. The four affine logarithms comprising F_0 make it a standard self-concordant barrier with parameter at most 4. Its Hessian is

$$H_0 = \text{Diag}(A_0, B_0), \quad A_0 = \frac{2(1 + x^2)}{(1 - x^2)^2} \geq 2, \quad B_0 = y^{-2} + (1 - y)^{-2} \geq y^{-2}.$$

For $h = (u, v)$ put $U = \sqrt{A_0}u$, $V = \sqrt{B_0}v$, so that $\|h\|_{H_0}^2 = U^2 + V^2$, $|u| \leq |U|/\sqrt{2}$, and $|v|/y \leq |V|$. Writing $s = \sin(\log y)$ and $t = \cos(\log y)$, direct differentiation gives

$$\begin{aligned} D\psi[h] &= su + xtv/y, \\ D^2\psi[h, h] &= 2tuv/y - x(s+t)v^2/y^2, \\ D^3\psi[h, h, h] &= -3(s+t)uv^2/y^2 + x(3s+t)v^3/y^3. \end{aligned}$$

Since $|x| < 1$, $|s+t| \leq \sqrt{2}$, and $|3s+t| \leq \sqrt{10}$, these imply

$$\begin{aligned} |D\psi[h]| &\leq 2\|h\|_{H_0}, \\ |D^2\psi[h, h]| &\leq 3\|h\|_{H_0}^2, \\ |D^3\psi[h, h, h]| &\leq 7\|h\|_{H_0}^3. \end{aligned} \tag{95}$$

For example the second bound follows from $\sqrt{2}|UV| + \sqrt{2}V^2 \leq 3(U^2 + V^2)$; the third follows from $3|U|V^2 + \sqrt{10}|V|^3 \leq 7(U^2 + V^2)^{3/2}$. For $f_0 = F_0 + \varepsilon\psi$ we therefore have

$$D^2f_0[h, h] \geq (1 - 3\varepsilon)\|h\|_{H_0}^2, \quad |D^3f_0[h, h, h]| \leq (2 + 7\varepsilon)\|h\|_{H_0}^3, \quad |Df_0[h]| \leq (2 + 2\varepsilon)\|h\|_{H_0}.$$

The inequality $2 + 7\varepsilon \leq 4(1 - 3\varepsilon)^{3/2}$ proves standard self-concordance of $F = 4f_0$. Its gradient parameter is at most

$$\frac{4(2 + 2\varepsilon)^2}{1 - 3\varepsilon} < 20. \tag{96}$$

Its Hessian is positive definite. The bounded perturbation ψ does not affect divergence at the boundary, completing the global check.

At $y = g_k$, $s = 0$ and $t = 1$. At $x = 0$ the first derivative in x vanishes, while

$$F_y(0, g_k) = 4\left(-\frac{1}{g_k} + \frac{1}{1 - g_k}\right) < 0.$$

Consequently $\mu_k = -1/F_y(0, g_k) > 0$ makes the point stationary for $F + y/\mu_k$. Strict convexity makes it the unique exact center. The objective gap is g_k . Define $J = H/4$ to keep the scaling explicit:

$$J = \begin{pmatrix} 2 & \varepsilon/g_k \\ \varepsilon/g_k & g_k^{-2} + (1 - g_k)^{-2} \end{pmatrix}. \tag{97}$$

For $g = g_k \downarrow 0$, its eigenvalues satisfy $\lambda_+(J) \sim g^{-2}$ and $\lambda_-(J) \rightarrow 2 - \varepsilon^2$, as follows from its trace and determinant. A unit weak eigenvector v_- with positive first coordinate obeys

$$\frac{(v_-)_2}{(v_-)_1} = -\frac{2 - \lambda_-(J)}{\varepsilon/g} \sim -\varepsilon g.$$

It follows that $|v_-^\top r| \sim \varepsilon g$, whereas $|v_+^\top r| \rightarrow 1$. Restoring $H = 4J$ gives

$$\|H^{-1}\Pi_F r\| \sim \frac{\varepsilon g}{4(2 - \varepsilon^2)}, \quad \|H^{-1}(I - \Pi_F)r\| \sim \frac{g^2}{4}. \tag{98}$$

The weak component thus dominates the Euclidean solution norm, while the residual of the strong-only solution is exactly $\Pi_F r$. This proves (94). $\square$

This example also explains why the general-barrier theorem does not assume convergence to one point of the optimal face. At every sufficiently small y , the unique solution of

$$\frac{2x}{1-x^2} + \varepsilon \sin(\log y) = 0$$

has $F_y(x, y) < 0$ and hence lies on the exact path for a positive μ . Indeed $F_y/4 \leq (-1+\varepsilon)/y + 1/(1-y) < 0$ for small y . The solution x takes distinct limiting values along sequences with $\sin(\log y) = 1$ and -1 . Thus the path has no single endpoint, despite the barrier having a finite parameter. The weak eigenspace still converges to the optimal-face tangent as the theorem requires.

Nonconvergent central paths were already constructed by [Bolte and Pauwels \(2022\)](#) using a Legendre function that is finite and continuous on a closed square. That function does not satisfy the boundary-divergence requirement in [definition 2.1](#). The example above establishes nonconvergence within the finite-parameter self-concordant class, with a global parameter certificate and the sharp projector and solution-error estimates proved here.

9 What spectral clustering implies for linear solves

Large spectral condition numbers need not force the usual $O(\sqrt{\kappa})$ conjugate-gradient iteration bound to be sharp. Classical cluster-sensitive estimates exploit the spectrum more precisely; see [Axelsson \(1994\)](#). We give the elementary polynomial construction needed for [theorem 8.1](#). This is an exact-arithmetic statement about energy-norm error, not a finite-precision residual theorem.

Proposition 9.1 (An explicit two-interval CG bound). *Suppose $H \succ 0$ and*

$$\text{spec}(H) \subseteq [a_1, b_1] \cup [a_2, b_2], \quad 0 < a_1 \leq b_1 < a_2 \leq b_2.$$

Put $\rho_i = b_i/a_i$ and $\Lambda = b_2/a_1$. For $0 < \delta \leq 1/2$ there is a real polynomial q , with $q(0) = 1$, such that $\max_{\text{spec}(H)} |q| \leq \delta$ and

$$\deg q = O(\sqrt{\rho_1 \rho_2} \log(4\Lambda) \log(1/\delta) + \sqrt{\rho_2} \log(1/\delta)). \quad (99)$$

Exact-arithmetic CG for $Hu = r$, from any initial vector u_0 , therefore achieves $\|u - u_k\|_H \leq \delta \|u - u_0\|_H$ within this many iterations, or within $\dim H$ iterations if that is smaller.

Proof. For $0 < a < b$ the normalized Chebyshev polynomial

$$r_m(x; a, b) = \frac{T_m((a+b-2x)/(b-a))}{T_m((a+b)/(b-a))}$$

has value one at zero and satisfies

$$\max_{[a,b]} |r_m| \leq 2 \left(\frac{\sqrt{b/a} - 1}{\sqrt{b/a} + 1} \right)^m \leq 2 \exp \left(-\frac{2m}{\sqrt{b/a}} \right). \quad (100)$$

The first bound follows from $|T_m| \leq 1$ on $[-1, 1]$ and $T_m(t) = (w^m + w^{-m})/2$ for $t = (w + w^{-1})/2 > 1$. The second uses $\log((s+1)/(s-1)) \geq 2/s$ for $s > 1$.

Choose $p_2 = r_{m_2}(\cdot; a_2, b_2)$ with $m_2 = \lceil \sqrt{\rho_2} \rceil$. Its magnitude is at most $1/2$ on the upper interval, and at most one on $[0, a_2]$, because $T_{m_2}(t)$ is positive and increasing for $t \geq 1$. If $a_2 = b_2$, use $p_2(x) = 1 - x/a_2$, $m_2 = 1$ instead. Choose $r_1 = r_{m_1}(\cdot; a_1, b_1)$ with $m_1 = \lceil \frac{1}{2} \sqrt{\rho_1} \log(2/\delta) \rceil$, so that $|r_1| \leq \delta$ on the lower interval. If $a_1 = b_1$, use $r_1(x) = 1 - x/a_1$ and $m_1 = 1$.

All m_1 roots θ_j of r_1 lie in $[a_1, b_1]$, and $r_1(0) = 1$, so its product representation gives

$$|r_1(x)| = \prod_{j=1}^{m_1} |1 - x/\theta_j| \leq \Lambda^{m_1} \quad (a_2 \leq x \leq b_2).$$

Let

$$j = \lceil m_1 \log_2(4\Lambda) + \log_2(1/\delta) \rceil, \quad q = r_1 p_2^j.$$

On the lower interval $|q| \leq \delta$; on the upper interval $|q| \leq \Lambda^{m_1} 2^{-j} \leq \delta$. Its degree $m_1 + jm_2$ gives (99). Finally, exact-arithmetic CG minimizes the energy-norm error over vectors $u - u_k = q_k(H)(u - u_0)$ with $q_k(0) = 1$ and degree at most k . The spectral theorem bounds the relative energy error of our candidate by $\max_{\text{spec}(H)} |q|$. The usual exact termination in dimension $\dim H$ gives the stated alternative. $\square$

For each fixed barrier in [theorem 8.1](#), the two interval ratios are bounded and $\Lambda = \Theta(g^{-2})$ when $f > 0$. Thus [proposition 9.1](#) gives $O(\log(1/g) \log(1/\delta))$ iterations on a sufficiently small tail. If $f = 0$, there is one interval with bounded condition number, and the ordinary Chebyshev estimate gives $O(\log(1/\delta))$. These are matrix-vector-product counts. They do not include the cost of producing those products, and floating-point loss of conjugacy can affect both attainable error and iteration counts. A recursively updated residual should therefore be checked against a freshly computed $r - Hu_k$ in any numerical use of the bound.

9.1 Quantum access and output are separate assumptions

A quantum linear-system algorithm commonly receives a unitary whose specified block equals H/α , up to a controlled encoding error, where $\alpha \geq \|H\|$ is its normalization. This is called a *block encoding*. The inverse scale seen by that encoding is

$$\kappa_{\text{enc}} = \alpha \|H^{-1}\|, \quad (101)$$

which can exceed $\kappa(H)$. Such an algorithm also needs a procedure preparing the normalized actual right-hand side and a specified output task, often preparation of the normalized solution rather than all its entries. These assumptions are part of the problem, not consequences of Hessian geometry.

Quantum singular value transformation implements suitably bounded polynomials of a normalized encoded matrix ([Gilyén et al., 2019](#)). The polynomial must meet the boundedness contract on the whole interval $[-1, 1]$, including points absent from the spectrum. In contrast, [proposition 9.1](#) controls its polynomial only on the two spectral intervals; its values in their intervening gap are unrestricted. That construction alone therefore supplies no quantum implementation with its CG degree as the query cost.

The distinction is consistent with the established lower bounds and structured positive results of [Orsucci and Dunjko \(2021\)](#). Their Proposition 6 proves a worst-case $\Omega(\min\{\kappa, N\})$ query lower bound for positive-definite quantum linear systems of dimension N , with the specified matrix and state-preparation oracles. The linear-in- κ regime consequently requires N at least of that order. Such a worst-case family is not by itself an accuracy lower bound along one fixed-dimensional central path. Their accelerated positive-definite constructions impose additional normalization, decomposition, or state-overlap conditions; merely writing a Gram factorization of H does not establish a generic square-root dependence on κ .

There is also an important right-hand-side restriction. The geometric theorems here concern all eigenvalues, whereas exact path-following forces the particular vector in (88). The window residual guarantee (91) can exploit that vector only under its stated projection and solve assumptions.

Moreover, [proposition 8.2](#) shows that a small discarded right-hand-side fraction can carry almost all of the Euclidean solution norm. Thus a residual guarantee cannot silently be substituted for a quantum solution-state guarantee.

Finally, Hessian conditioning is not an obstruction to every quantum approach to optimization. [Apers and Gribling \(2026\)](#) give quantum speedups for tall LPs with row access, using spectral approximations and gradient estimation to return an explicit feasible approximate solution. Their method avoids the Hessian-condition-number dependence of a direct quantum linear-system substitution. The results here describe a fixed metric and its spectrum; they neither rule out that algorithm nor prove an end-to-end classical–quantum runtime separation.

10 Metrics, formulations, and the finite-parameter boundary

10.1 Fixed coordinate changes and gap-dependent preconditioning

All preceding spectral rates use a fixed external metric on the feasible affine hull. If $R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a fixed invertible change of tangent coordinates, the Hessian matrix becomes $R^\top H R$. Using ordinary Euclidean matrix condition numbers in the respective coordinate systems gives

$$\frac{\kappa(H)}{\kappa(R)^2} \leq \kappa(R^\top H R) \leq \kappa(R)^2 \kappa(H). \quad (102)$$

The upper bound follows by bounding the extreme Rayleigh quotients using the extreme singular values of R ; the lower bound follows by applying the upper bound to R^{-1} . Thus fixed coordinate changes preserve accuracy exponents, with possibly large constant factors. If the original external metric is transported instead, its Gram matrix is $R^\top R$, and the generalized spectrum of $(R^\top H R, R^\top R)$ is exactly the spectrum of H . A gap-dependent transformation is different: the mathematical choice $R(g) = \hat{H}_F(g)^{-1/2}$ produces the identity as the Euclidean matrix for the transformed linear solve. Whether such a preconditioner is useful depends on its construction, application, and representation costs. Neither the geometric theorem nor [\(102\)](#) bounds those costs.

10.2 The reduced Hessian is not the Schur or augmented matrix

To distinguish the matrices, let $B = \nabla^2 F(x) \succ 0$ be an *ambient* barrier Hessian for equality constraints $Ax = b$. The reduced matrix is $W^\top B W$ for an orthonormal basis of $\ker A$; the primal Newton Schur matrix is $AB^{-1}A^\top$. The latter expression must use B , not the already reduced operator. Under an invertible ambient coordinate change $z = Gx$, transport the barrier as $\tilde{F}(z) = F(G^{-1}z)$ and put $\tilde{A} = AG^{-1}$. Then

$$\tilde{B} = G^{-\top} B G^{-1}, \quad \tilde{A} \tilde{B}^{-1} \tilde{A}^\top = AB^{-1}A^\top. \quad (103)$$

In contrast, the augmented Newton matrix transforms by congruence:

$$\begin{pmatrix} \tilde{B} & \tilde{A}^\top \\ \tilde{A} & 0 \end{pmatrix} = \begin{pmatrix} G^{-\top} & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} B & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} G^{-1} & 0 \\ 0 & I \end{pmatrix}. \quad (104)$$

Congruence is not spectral equality. In particular, a bound for the reduced primal Hessian cannot be read as the same bound for every Schur, scaled primal–dual, or augmented formulation. Classical conditioning analyses such as [Peña \(2002\)](#); [Renegar \(1996\)](#) must be compared with the actual matrix they study.

For an explicit illustration of (103), consider the Lorentz cone and its barrier $F(t, z) = -\log(t^2 - \|z\|^2)$. For $0 < \lambda < 1$, let G_λ act on the first two coordinates as

$$\frac{1}{2} \begin{pmatrix} \lambda + \lambda^{-1} & \lambda - \lambda^{-1} \\ \lambda - \lambda^{-1} & \lambda + \lambda^{-1} \end{pmatrix}$$

and as the identity on the other coordinates. It preserves $t^2 - \|z\|^2$ and the positive sheet, so $F(G_\lambda x) = F(x)$. At $e_0 = (1, 0, \dots, 0)$ the ambient Hessian is $2I$, while at $G_\lambda e_0$ it is $2G_\lambda^{-2}$, with condition number λ^{-4} . Nevertheless, transforming A as above leaves the Schur matrix exactly unchanged. This is a coordinate illustration on the cone, not an application of the compact-feasible-set theorem to that unbounded cone. A varying λ can also change coefficient magnitudes and encoding normalizations, even though the transformation mixes only two coordinates.

The intrinsic length of a differentiable curve, $\int \sqrt{\dot{x}^\top B(x) \dot{x}} dt$, is preserved by the same transported metric. Its geometric role is studied by [Nesterov and Todd \(2002\)](#). This integral and a pointwise external condition number describe different quantities. Sensitivity to the objective also requires specifying an absolute scale. At fixed μ , implicit differentiation of exact centrality gives

$$D_{c_V} x_F(\mu) = -\mu^{-1} H_F(x_F(\mu))^{-1}. \quad (105)$$

Thus a lower Hessian eigenvalue bound does control this absolute objective-to-center derivative; the condition number alone does not. A derivative with respect to a chosen family parameter additionally includes the derivative of the data. For example, choose a short line segment $z_0 + [-a, a]v$ contained in a fixed barrier domain and set $z_k(t) = z_0 + a \sin(kt)v$. Continuity and positive definiteness give uniform spectral bounds for the Hessian on that segment, while $\|z'_k(0)\| = ak\|v\|$ is unbounded. Taking $c_k(t) = -\nabla F(z_k(t))$ makes each point an exact center for $\mu = 1$ with that varying objective. The data derivatives vary too, consistently with (105). The example establishes a separation between pointwise spectral bounds and variation with respect to an unrestricted family parameter. It does not show unbounded sensitivity per unit objective perturbation when the Hessian is uniformly bounded below. More generally, differentiating an auxiliary representation of a central point introduces derivatives of that representation's data maps. Comparing its sensitivity with Newton conditioning requires bounds on those maps and on the Jacobian of its defining equations; pointwise centrality alone does not supply that comparison.

10.3 Why a finite barrier parameter is essential

The gradient inequality (7) is essential to the containment theorem. Differential self-concordance (6) and boundary divergence alone do not suffice. On the same rectangle as [proposition 8.2](#), for fixed $\eta > 0$ consider

$$F_\eta(x, y) = -\log(1 - x^2) - \log y - \log(1 - y) + \frac{\eta}{y}. \quad (106)$$

This function is convex, diverges at the boundary, and is differentially self-concordant. To check the last assertion, for $h(y) = -\log y + \eta/y$ write $t = \eta/y \geq 0$. Then

$$h''(y) = y^{-2}(1 + 2t), \quad |h'''(y)| = 2y^{-3}(1 + 3t), \quad (1 + 3t)^2 \leq (1 + 2t)^3,$$

where the difference of the right and left sides is $3t^2 + 8t^3$. Adding the other affine logarithms preserves differential self-concordance, by summing their third-derivative bounds and using $\sum a_i^{3/2} \leq (\sum a_i)^{3/2}$ for $a_i \geq 0$.

For objective y , its central tail has $x = 0$ and gap $g = y$. Indeed $\partial_x F_\eta(0, g) = 0$ and $\partial_y F_\eta(0, g) = -1/g + 1/(1 - g) - \eta/g^2 < 0$ for small g , so $\mu(g) = -1/\partial_y F_\eta(0, g) > 0$ makes $(0, g)$ the unique center by strict convexity. The diagonal Hessian has entries 2 and $g^{-2} + (1 - g)^{-2} + 2\eta g^{-3}$, so

$$\widehat{\kappa}_{F_\eta}(g) \sim \eta g^{-3}, \quad \frac{|\partial_y F_\eta(0, g)|^2}{\partial_{yy} F_\eta(0, g)} \sim \frac{\eta}{2g} \longrightarrow \infty. \quad (107)$$

Hence no finite ν satisfies the barrier-gradient inequality. This does not contradict the $\Theta(g^{-2})$ finite-parameter law for a positive-dimensional optimal face. The same distinction between differential self-concordance and a finite global gradient parameter is explicitly noted for inverse-power perturbations in [Monteiro and da Silva \(2026, Remark 11.1\)](#); no result of that preprint is needed for the calculation here.

11 Reproducible numerical illustrations

The computations check explicit formulas and illustrate finite-gap behavior. They do not estimate an unproved asymptotic exponent or benchmark an end-to-end interior-point algorithm. The accompanying reproduction package contains all input arrays, exact rational certificates, scripts, raw rows including rejected points, generated figures and tables, and a manifest of commands, software versions, random seeds, transformations, and file hashes. Regeneration uses Python, NumPy, SciPy, and Matplotlib without network access or imports from another repository. High-precision scalar calculations and reference solves use the standard-library `decimal` module.

11.1 Exact examples

[Figure 1](#) evaluates the exact central-point and stable quadratic-root formulas from [theorem 7.2](#) with 80 decimal digits, for $g = 10^{-2}, 10^{-3}, \dots, 10^{-24}$. Dividing the four scaled eigenvalues by their proved limits displays convergence to one. The paired section retains the two (b, g) eigenvalues and consequently has a different condition-number exponent. At $g = 10^{-2}, 10^{-4}, 10^{-6}$, a separate float64 computation forms the Hessian through $\text{tr}(X^{-1}UX^{-1}V)$ and solves the generalized eigenproblem using the tangent Gram matrix. Its eigenvalues agree with the scalar formulas to relative error below 10^{-7} . The high-precision formulas avoid subtracting nearly equal quadratic roots by obtaining the smaller root from the determinant divided by the larger root.

For the simplex of [proposition 5.4](#), the exact log-barrier center satisfies $x_i = \mu/(c_i + z)$, where $z > 0$ is the unique root of $\sum_i \mu/(c_i + z) = 1$. We solve this scalar monotone equation with 70 decimal digits and evaluate the Hessian through its factor. [Figure 2](#) shows the inverse-square regime giving way to the large limiting plateau when g falls below ϑ . For the compact degenerate LP of [example 6.2](#), direct centering gives, at $\mu = 10^{-9}$, $\mu^2 \lambda_{\min} \approx 0.24888160$ and $\mu^2 \lambda_{\max} \approx 1.14289068$, consistent with the proved limit and $\kappa(L) \approx 4.59210571$. The full sequence and its centrality residuals are supplied in the data package.

[Table 1](#) evaluates the exact subsequence in [proposition 8.2](#). It records the norm of the discarded forcing component divided by g , and the relative Euclidean error of the solution obtained by retaining only the strong eigenmode. The forcing component tends to zero at the proved rate while the solution error tends to one. This is a direct check of the distinct residual and solution contracts, not an iterative-solver artifact.

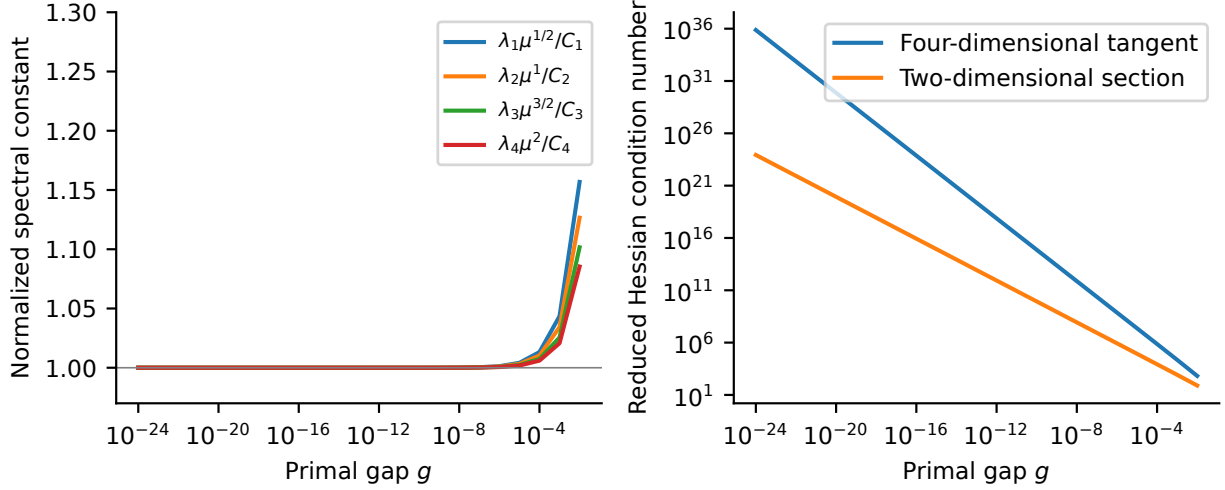


Figure 1: The fractional SDP and its section. Left: the scaled ordered eigenvalues divided by $C = (\sqrt{2}, 1, \sqrt{2}, 4/7)$, with powers $(1/2, 1, 3/2, 2)$ of μ . Right: the condition numbers on the two tangent spaces. Both optimality systems have singularity degree two.

k	g_k	$\ \Pi c\ /(g_k\ c\)$	Relative solution error
1	1.87e-03	0.01000003	0.93680040
2	3.49e-06	0.01000000	0.99999976
3	6.51e-09	0.01000000	1.00000000
4	1.22e-11	0.01000000	1.00000000

Table 1: Oscillatory barrier with $\epsilon = 0.01$ and $g_k = \exp(-2\pi k)$. The relative solution error is $\|H^{-1}\Pi c\|/\|H^{-1}c\|$. Rounded values of one do not assert exact equality at finite gap.

11.2 Certified benchmark representations and precision screens

We use three small Netlib LPs (Gay, 2013): **afiro**, **sc50b**, and **adlittle**. The frozen arrays are the result of HiGHS 1.15.1 default presolve followed by conversion of inequalities and finite bounds to equality constraints and nonnegative variables. Their dimensions, from original MPS to presolved model to standard form, are respectively $(27, 32) \rightarrow (7, 10) \rightarrow (9, 18)$, $(50, 48) \rightarrow (15, 15) \rightarrow (15, 29)$, and $(56, 97) \rightarrow (53, 95) \rightarrow (55, 136)$. The Hessian uses the Euclidean metric in these final coordinates. It is not the Hessian of the original MPS representation. The frozen arrays define the tested instances, so a later presolver version is not needed for reproduction.

The hypotheses are checked without feasibility tolerances. Every stored float64 input is interpreted as its exact binary rational. The package verifies a rational point $x > 0$ with $Ax = b$ and a rational vector y with

$$A^\top y + \eta c > 0, \quad \eta \geq 0.$$

For $x \geq 0$ in a fixed upper objective sublevel, $x^\top (A^\top y + \eta c) = b^\top y + \eta c^\top x$ bounds every coordinate. The certificates use $\eta = 0$ for **afiro** and **sc50b**, proving compactness of their feasible sets, and $\eta = 1$ for **adlittle**, proving compactness of its objective sublevels. Thus the localized theorem applies to the last instance. Rational primal and dual feasible points also certify brackets on the optimal value, with widths approximately $8.2 \cdot 10^{-12}$, $9.4 \cdot 10^{-12}$, and $4.5 \cdot 10^{-7}$, respectively.

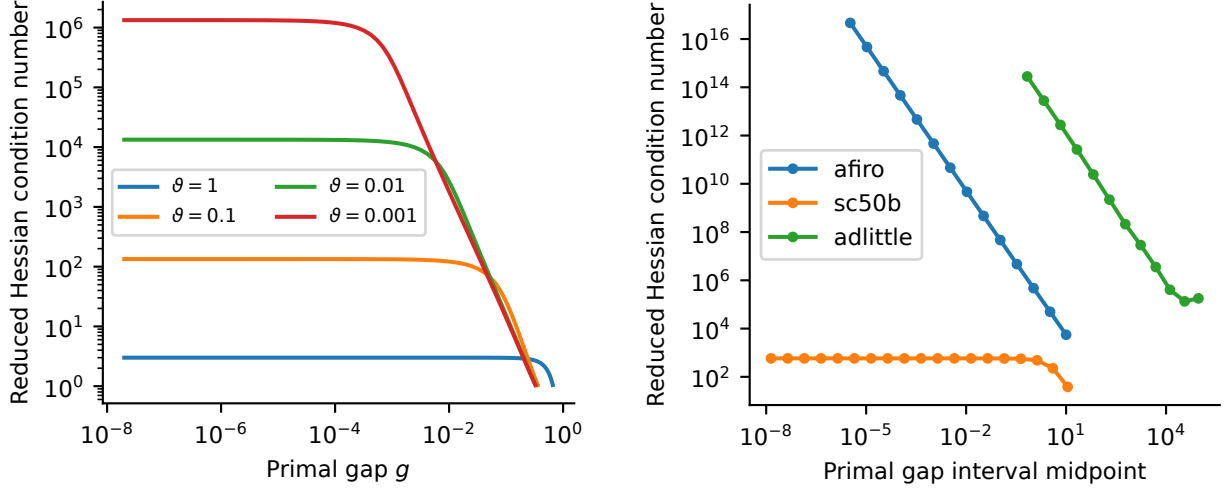


Figure 2: Left: near-degenerate simplex plateaus for fixed values of ϑ . Right: accepted finite-gap points for the frozen Netlib representations. The latter curves are numerical observations; no asymptotic exponent is fitted to them.

Instance	(m, n)	Accepted/attempted	Last gap	κ	ρ
afiro	(9, 18)	14/27	3.34e-06	4.65e+16	2.2e-07
sc50b	(15, 29)	19/27	1.40e-08	5.82e+02	6.1e-07
adliddle	(55, 136)	12/27	6.80e-01	2.81e+14	2.5e-09

Table 2: Accepted Netlib points and the last accepted point in each continuation sequence. All 27 attempted points per instance, including centrality, gap, repair, or spectral-resolution rejections, are retained in the raw data.

Centering uses an orthonormal nullspace basis W and the Hessian factor $B = \text{Diag}(x^{-1})W$. Singular values of B give the Hessian eigenvalues by squaring; Newton solves and decrements use the same factorization. Each stored iterate is repaired to an exactly feasible rational point using a fixed nonsingular column basis. We evaluate its objective exactly and subtract the certified optimal-value bracket. A point is accepted for the plot only when this gap interval is positive and has relative width at most 1%, the repaired point is positive, the maximum relative coordinate repair is at most 10^{-4} , the scaled equality residual is at most 10^{-12} , and the *unscaled* centrality decrement is at most 10^{-5} . We additionally require

$$\epsilon_{\text{mach}} \max(n, d) \kappa_2(B) \leq 10^{-6}.$$

This last screen is a numerical resolution indicator for the factor SVD, not a rigorous interval certificate for its singular values. The exact rational certificates establish the instance hypotheses; the plotted spectral values remain finite-precision computations.

Table 2 makes the different resolved accuracy ranges explicit. In particular, the stringent spectral screen stops *adliddle* well before its deepest attempted gap. We draw no conclusion about its limiting condition number from that truncation. The near-degenerate simplex already proves why a finite observed slope or plateau should not by itself be treated as an asymptotic classification.

g	Iterations	Recursive residual	True residual	Relative energy error
1e-01	43	8.82e-09	8.82e-09	3.54e-09
1e-02	71	2.56e-09	2.56e-09	4.21e-09
1e-03	96	4.37e-09	4.39e-09	4.24e-09
1e-04	124	2.95e-09	2.52e-08	8.55e-09

Table 3: Float64 CG on the synthetic two-interval matrices. The final two columns use an 80-digit reference calculation for the actual rounded matrix and right-hand side. The last recursive stopping test does not certify a true relative residual below 10^{-8} .

11.3 Clustered CG and the meaning of stopping accuracy

To exercise two intervals with many distinct eigenvalues, we generate a 60×60 symmetric matrix from a fixed-seed orthogonal rotation of 30 equally spaced values in $[1, 2]$ and 30 in $[g^{-2}, 3g^{-2}]$. This is a synthetic spectral test, not a claimed Newton Hessian. The actual float64 matrix is explicitly symmetrized. CG uses ordinary float64 matrix-vector products and stops when its recursively updated relative residual first reaches 10^{-8} . At that iterate, an 80-digit solve of the exact real system defined by the stored float64 matrix and right-hand side supplies the reference solution, relative energy error, and independently evaluated true residual.

The prescribed internal interval ratios are two and three while the interval separation grows; these ratios precede floating-point matrix construction and are not exact spectral assertions for the rounded matrix. Table 3 illustrates the benefit of this structure and the distinction between stopping metrics. It is not an empirical proof of the exact-arithmetic degree bound. Indeed, exact-arithmetic CG terminates in at most 60 steps for these matrices, whereas floating arithmetic can take more steps because conjugacy is lost. Recomputed residuals and energy error therefore remain separate reported quantities, as required by the solver contract in section 9.

12 Discussion

The central-path Hessian is constrained by an object that does not refer to the barrier: the difference body of the current objective sublevel. This explains both the matching condition-number law and the comparison of complete spectra at equal accuracy. Within a fixed representation and metric, a finite-parameter barrier change cannot remove a conditioning exponent imposed by sublevel geometry. Uniform bounds on barrier parameters are essential if the barrier itself varies with the requested gap.

The resulting geometric distinction is finer than several customary classifications. A unique optimum does not by itself imply bounded conditioning outside the polyhedral setting. Conversely, primal vertex degeneracy does not prevent a positive-definite scaled log-Hessian limit in an LP. In SDP, an attained fractional error bound can determine the conditioning exponent, but singularity degree alone cannot: the paired degree-two systems have different diameter rates. Full-spectrum width bounds explain why an extreme-eigenvalue ratio also need not describe the interior spectral scales.

For linear solves, the direction of the forcing and the accuracy contract remain decisive. On an LP central tail, the weak eigenspace approaches the optimal-face tangent and the exact Newton forcing has a small projection onto it. The sharp barrier example shows why this fact is insufficient for accurate direction recovery after discarding weak modes. Clustered CG can nevertheless exploit separated spectral intervals in exact arithmetic. Its conclusion is an energy-error bound, and the numerical examples expose the need to distinguish it from a recursively computed residual. Quantum

applications require additional access and output assumptions before either fact can be converted into a runtime statement.

These are statements about a specified primal operator. They coexist with useful changes of representation, gap-dependent preconditioners, and primal–dual formulations whose spectra can behave differently. The precise comparison established here supplies a common geometric reference for assessing such changes: it identifies what is forced by objective sublevels before accounting for the cost and effect of the linear algebra used to solve the Newton equations.

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